

Predictive latent dynamics for parametric vortex-gust airfoil interactions: a JEPA benchmark against reconstructive and linear baselines

Carlos Sanmiguel Vila^{1,2}

¹INTA, Instituto Nacional de Técnica Aeroespacial

²Universidad Carlos III de Madrid

Abstract

We test whether a predictive latent training objective produces a reduced-order representation of a parametric gust-vortex airfoil interaction that is more useful for forward-time forecasting than a reconstructive one. The dataset is a direct numerical simulation of a NACA 0012 airfoil at angle of attack $\alpha = 14^\circ$ and chord Reynolds number $Re = 5000$ perturbed by Taylor vortices parametrised by gust strength G , vortex core diameter D , and wall-normal offset Y , with 84 cases split into 226 training, 28 validation, 42 stratified test, and 24 out-of-distribution extrapolation encounters. A Joint Embedding Predictive Architecture is compared head-to-head with the Fukami observable-augmented autoencoder and a proper orthogonal decomposition basis at matched latent dimensions $d \in \{16, 32, 64\}$ under a fairness protocol in which each baseline shares an identical autoregressive transformer predictor, the identical training recipe, the identical conditioning on (G, D, Y) , and the identical probe family; the encoder is the only thing that varies. The headline metric is the closure of the predictor rollout to six physical observables at horizon $H = 16$. The predictive latent achieves a mean coefficient of determination of 0.835 at $d = 64$, against 0.561 for the linear basis and 0.427 for the reconstructive autoencoder at the same dimension; the gap is largest on the spatially distributed wake observables. The matched-capacity result at $d = 32$ retains a mean R^2 of 0.808. The mechanism behind the gap is a latent drift diagnostic that shows the reconstructive autoencoder rollout goes one order of magnitude out of distribution at $H = 16$ while the predictive latent stays inside its training manifold. A conditioning-only floor establishes that the generalisation gap to the autoencoder is not explained by the predictor’s parametric conditioning. The predictor is the latent dynamics function required by model-based control and the architectural pathway to a closed-loop deployment is sketched.

1 Introduction

1.1 Unsteady gust-airfoil interactions and the case for reduced-order models

Sharp transverse gusts interacting with lifting surfaces at low chord Reynolds numbers are a routine operating condition for fixed-wing micro air vehicles, urban-air-mobility rotorcraft, and small uncrewed aircraft flying in the atmospheric boundary layer or in the wake of buildings and vehicles. The unsteady aerodynamic response is dominated by the timing of the vortex impact relative to the airfoil’s own shedding cycle, by the wall-normal offset of the vortex core relative to the leading edge, and by the gust strength and core diameter. Even at the moderate Reynolds number of $Re=5000$ the post-stall flow at angle of attack $\alpha=14$ degrees is fully separated, and the gust- induced disturbance interacts non-linearly with the natural shedding [9, 22]. Direct numerical simulation (DNS) of the full parametric envelope is expensive enough

that reduced-order modelling (ROM) of the encounter-to-encounter and case-to-case variability is an active area of work.

Recent ROM efforts on this regime have converged on observable- augmented autoencoder architectures. Fukami and Taira’s lift-augmented autoencoder [8] trained an encoder, a decoder, and a CL prediction head jointly on transonic airfoil flow and reported that the auxiliary lift-prediction loss reduced the intrinsic dimension of the discovered manifold from approximately ten to three. Solera-Rico et al. [22] (Nat. Commun. 2024) used a beta-VAE plus transformer ROM conditioned on aerodynamic observables to forecast latent dynamics on a vortex-gust airfoil dataset of the same family as the data in this paper. Fukami et al.’s 2025 transonic-buffet extension [9] (J. Fluid Mech. 2025) generalised the observable-augmentation idea across multi-source turbulent flow data. The common ingredient is a reconstruction-trained encoder that also predicts an aerodynamic scalar (CL, drag, surface pressure) from the bottleneck. The reconstruction term anchors the latent geometry; the observable prediction term shapes the manifold along the directions that matter for downstream forecasting.

What has not been tried on this regime is a Joint-Embedding Predictive Architecture (JEPA): a class of self-supervised latent dynamics models that drops the reconstruction term entirely and trains the encoder and predictor end-to-end only on latent-space forecasting plus an anti-collapse regulariser. Removing the reconstruction term frees the encoder from spending capacity on photorealistic detail and lets it focus on the dynamically relevant degrees of freedom. The JEPA lineage runs from the I-JEPA images formulation through V-JEPA on video [1], with two recent end-to-end-from-pixels variants that close the loop on a moving target encoder: LeWM [15] (Maes et al., arXiv:2603.19312, 2026), which trains a single online encoder against the characteristic-function SIGReg regulariser of LeJEPA [3] (Balestrierio and LeCun, arXiv:2511.08544, 2025), and PLDM [21] (Sobal et al., arXiv:2502.14819, 2025), which uses a five-term VICReg-derived objective [5]. Both architectures have so far been benchmarked on gridworld and toy visual domains; their behaviour on a low-intrinsic-dim physics dataset, where the data itself sits on a roughly five- to ten- dimensional manifold, is not characterised.

1.2 What this paper contributes

We train an end-to-end JEPA on DNS of a NACA 0012 airfoil at $Re=5000$, $\alpha=14$ degrees, perturbed by a parametric Taylor vortex characterised by gust strength G , vortex core diameter D , and wall-normal offset Y/c . The data partition v1.2 holds 41 train cases (138 encounters), 6 Test B cases at unseen interior (G, D, Y) values (28 encounters), and 4 Test C cases at the $|G|=4$ extrapolation boundary (24 encounters); split sha256 a721dc92f6e278ee054bb952933c14ba20a58137f79f3a19fc6ad71 locked at `configs/splits/split_v1.json`. The encoder is a hybrid CNN+ViT mapping mid-plane spanwise vorticity (192×96) to a $d=32$ latent through a one-layer MLP projection with BatchNorm [15]; the predictor is a six-layer autoregressive transformer with rotary positional embeddings [23] and AdaLN-Zero conditioning on $c=(G, D, Y)$. Section 3 details the architecture, the two-term SIGReg [3] and five-term PLDM [21] loss compositions, and the optional auxiliary CL-prediction head we adopt from the Fukami / Solera-Rico observable-augmented lineage. The headline figure of merit on each held-out split is $\text{delta} = r2(z \rightarrow \text{CL_future}) - r2((c, t) \rightarrow \text{CL_future})$, the gain over a tiny parametric MLP baseline that maps (`case_descriptor`, `time_index`) to CL.

The paper’s three contribution claims, with their headline numbers from Sessions 7 and 8 (see Section 5):

Contribution 1 (controlled-collapse SIGReg + OBS at scale). The production cell at the ($\eta \times \lambda$) grid optimum ($\eta=0.01$, $\lambda=0.01$, $d=32$) reaches mean Test B $\text{delta} = +0.131 \pm 0.032$ (1-sigma) across three random seeds over the parametric baseline at 20k training iterations (Session 8 D53 E4 + Session 9 D58 F4 + F5). This is the simpler of the two anti-collapse bases in the JEPA literature (two-term SIGReg, not five-term PLDM) augmented by a small CL-prediction head. The encoder converges into a controlled-collapse regime: $PR(z)$

stays in the 2 to 3 range (PR_all between 2.61 and 3.10 on Test B across the three production seeds, versus $d = 32$) while the latent retains Test A to Test B alignment strong enough to drive a CL prediction r^2 of approximately 0.82 to 0.88 on unseen parametric cases. The same configuration extrapolates to Test C at $|G| = 4$ with mean $\Delta = +0.474$ across seeds (Section 5.6 Table 3).

Contribution 2 (the observable head is load-bearing for both anti-collapse bases at full scale). Pure SIGReg + BatchNorm without the observable head (R0 control, D55) fails catastrophically on Test B at both $\lambda=0.1$ and $\lambda=0.01$: $\Delta_{\text{test}_b} = -0.74$ and -0.75 respectively. Pure five-term PLDM without the observable head (R2, D47) fails at $\Delta_{\text{test}_b} = -0.85$. Adding the observable head at $\eta = 0.01$ contributes $+0.87$ absolute Test B Δ to the SIGReg path (R0 -0.74 to the 3-seed-mean E4/F4/F5 $+0.131$) and $+0.84$ absolute to the PLDM path (R2 -0.85 to R1 -0.003). The two regulariser bases are similar in their dependence on the observable head; neither produces a generalising latent without it at this data scale.

Contribution 3 (regulariser-asymmetry inversion at scale, with a controlled-collapse mechanism). At the 5-case smoke scale the five-term PLDM + observable head reached the HEALTHY quadrant of the static-versus-dynamic diagnostic with PR around 10 while two-term SIGReg + observable head plateaued at PR around 3 (D39, Section 4). The smoke-scale reading was "PLDM is the recommended base." At the full 41-case scale the reading inverts. SIGReg + observable head (R3, the Session 7 anchor) generalises to Test B at $\Delta = +0.14$; PLDM + observable head (R1) does not ($\Delta = -0.003$). The asymmetry is robust to PLDM hyperparameter choice: a PLDM run with the paper-tuned Two-Rooms weights from [21] Appendix J.2 ($\alpha=4.0$, $\beta=6.9$, $\delta=0.75$, $\omega=0.0$) is even worse (E10, D53b: $\Delta_{\text{test}_b} = -0.095$). The participation ratio that looked healthy on PLDM at the smoke scale, where the 5-term loss has only 5 case labels to memorise, encodes case-specific memorisation once 41 cases are available; PR_all rises in lockstep with cross-split degradation in R2's trajectory audit, while PR_within shrinks (D50). The mechanism behind the SIGReg + OBS win is a controlled-collapse equilibrium: at $\lambda=0.01$ the SIGReg loss provides directional pressure that prevents rank-1 collapse without forcing the encoder to maintain the high-PR isotropy that pure SIGReg would impose. The PLDM five-term loss cannot be "bought off" cheaply because all four collapse-prevention terms pull simultaneously, and the resulting full-scale latent absorbs case-axis variation that does not transfer to Test B. The LeWM Two-Room intrinsic-dimension prediction (smaller d closer to the data's intrinsic dim should win) is not confirmed on this data either: the d sweep at the production operating point (d in 8, 16, 32) finds $d^*=32$ best on Test B by $+0.07$ over $d=8$ (D54), with PR_all flat across d . The extra dimensions help the downstream cross-split probe, not the encoder.

The Test B Δ over the (c, t) parametric baseline is the diagnostic that survives the smoke-to-full-scale inversion. The participation ratio alone, the headline number in the SIGReg [3] and PLDM [21] literature, is anti-correlated with generalisation on R2's trajectory and does not separate "encoder memorised 41 case labels" from "encoder learned dynamics" in this regime. The static-versus-dynamic decomposition introduced in Section 4, combined with the (c, t) baseline, gives a more robust audit.

1.3 Honest scope

The paper reports a single locked partition (v1.2), a single training duration (20k iterations), and a single-seed grid sweep with a single-seed-variance datapoint at the R3 anchor (seed=42, Test B $\Delta = +0.121$; pass criterion $[+0.05, +0.25]$, D52). Section 5.9 lists these and the other limitations of the evidence base. The λ bisection over the finer interval around $\lambda^*=0.01$ is running concurrently with the writing of this manuscript; the visualisation decoder on the frozen $d=32$ SIGReg + OBS encoder is also a Session 9 deliverable. The numbers in Section 5 are the Session 7 and 8 production runs only; subsequent revisions will fold in the bisection result and the decoder reconstruction metrics.

1.4 Roadmap

Section 2 reviews related work on observable-augmented autoencoders for unsteady aerodynamics, on JEPA architectures, and on anti-collapse regularisation for self-supervised representation learning. Section 3 specifies the data partition, the encoder and predictor architectures, the two-term SIGReg and five-term PLDM loss compositions, the observable head, and the diagnostic suite. Section 4 walks through the four failure modes that pure two-term SIGReg JEPA hit at the 5-case smoke scale, organised as the TRIVIAL / SPREAD_TRIVIAL / HEALTHY 2x2 of latent rank versus case-versus-dynamics encoding, and introduces the observable-augmentation rescue. Section 5 reports the full-scale results: the Session 7 three-way comparison (R1 PLDM+OBS, R2 PLDM-only, R3 SIGReg+OBS), the Session 8 validation diagnostics (trajectory audit, head ablation, seed-variance bound), the ($\eta \times \lambda$) grid that locates the production operating point, the latent-dimension sweep, and the R0 control that confirms the observable head is load-bearing. Section 6 reports the visualisation decoder trained on the frozen SIGReg + OBS encoder. Section 7 runs the architecture-spec ablation suite. Section 8 concludes.

2 Related work

2.1 Joint-embedding predictive architectures (JEPAs)

The architectural family we work in was outlined by LeCun in "A Path Towards Autonomous Machine Intelligence" ([13], 2022) as joint-embedding predictive architectures: an encoder maps an input x to a representation z , a predictor takes a context z_{ctx} and an action or temporal increment a and produces a target representation z_{tgt} , and the training objective minimises the distance between the predicted and the actually observed target embedding. The key contrast with autoencoders or pixel-space world models is that the predictor operates in latent space and the loss never reconstructs pixels. Two mechanical consequences follow that the present paper inherits. First, the encoder is free to discard information that is not predictive of future latents; second, the predictor cannot "cheat" via pixel-level shortcuts because no pixel-level signal is available to it. The cost is that the encoder can collapse to a constant representation that is trivially predictable, which is why every JEPA in the literature pairs the predictive loss with an explicit or implicit anti-collapse mechanism.

The first scaled instantiation in the image domain is I-JEPA (Assran, Duval, Misra, Bordes, Vincent, Morcos, Ballas, LeCun, CVPR 2023; [2]), which predicts masked-image patches in latent space from a context patch using a ViT encoder, predictor, and a target encoder updated via EMA. I-JEPA established that masked-modelling without pixel reconstruction is competitive on standard image classification probes and that the predictor must operate on positional information about the masked regions so the encoder is not penalised for losing absolute position. We do not use masking in this paper because temporal continuity of the vorticity field already supplies the predictive supervision signal, but the masked-image precedent motivates our design choice that the encoder be unconditional and the predictor be the conditioned component.

V-JEPA (Bardes, Garrido, Ponce, Rabbat, LeCun, Bojanowski, Ballas, arXiv:2404.08471, 2024; [4]) extended I-JEPA from images to short video clips and showed that a single-step latent prediction objective produces useful representations for action recognition. The follow-up V-JEPA 2 and V-JEPA 2-AC (Assran et al., arXiv:2506.09985, 2025; [1]) generalised the recipe to longer rollouts with scheduled sampling: a multi-step open-loop rollout where the predictor is conditioned on its own previous prediction with probability p that ramps up during training. We adopt exactly the V-JEPA 2-AC scheduled- sampling design ($H_{\text{roll}} = 8$, p linearly ramped from 0 to 1 over the first half of training; D21 in HANDOFF.md) because it is the cleanest published recipe for training a predictor to be stable under its own errors. The vanilla V-JEPA stop-gradient and EMA target encoder are intentionally omitted here following the LeWM thesis (Section 2.1 below) that an explicit characteristic-function regulariser obviates the need for

either.

LeWM (Maes, Le Lidec, Scieur, LeCun, Balestrieri, "LeWorldModel: Stable End-to-End Joint-Embedding Predictive Architecture from Pixels," arXiv:2603.19312, March 2026; [15]) is the architectural template the present work follows most closely. LeWM removes both stop-gradient and EMA by replacing them with two ingredients: a BatchNorm-projected encoder output and a characteristic-function regulariser (SIGReg) applied to the joint batch of all encoded frames in a sub-trajectory. The combination is shown to train stably from pixels on the Two-Room gridworld and the DMC Walker / Cheetah / Quadruped suites at sample efficiencies comparable to model-based RL, without any contrastive pairs and without an auxiliary reconstruction loss. Two LeWM appendices are directly load-bearing for our work. Appendix A gives the SIGReg mechanics (M projections, Epps-Pulley test against a Gaussian target distribution, BatchNorm at the latent boundary). Appendix G specifies the lambda bisection used to set the regulariser weight: six to eight evaluations over a logarithmic interval, with each evaluation a full training run, picking the lambda that maximises a downstream probe metric. We use this exact protocol in Session 9 over the lambda interval centred on the ($\eta \times \lambda$) grid optimum identified in Section 5.5. The LeWM Two-Room results also include the observation that SIGReg's effectiveness depends on the relationship between latent dimension d and the intrinsic dimension of the underlying data: when d is close to the intrinsic dim, SIGReg's anti-collapse pressure does not have to "fight itself" by spreading the latent uniformly across unused dimensions. Section 5.6 of the present work tests this LeWM prediction empirically on physics data and finds it does not transfer to the OBS-augmented regime we operate in (D54).

LeJEPA (Balestrieri and LeCun, "LeJEPA: Provable and Scalable Self- Supervised Learning Without the Heuristics," arXiv:2511.08544, November 2025; [3]) provides the formal SIGReg analysis on which the LeWM appendix-A implementation rests. The key results are two: an isotropic-Gaussian latent distribution is the maximum-entropy distribution under fixed second moments, so penalising deviations from Gaussian via a characteristic-function test is the distribution-free analogue of penalising deviations from white-noise covariance; and SIGReg is invariant to invertible affine reparametrisations of the latent, which removes the need for any covariance-shaping component analogous to VICReg's variance and covariance terms. The LeJEPA paper also gives the clean rationale for why BatchNorm rather than LayerNorm must be used at the latent boundary: SIGReg compares the empirical characteristic function of projections of z to that of a unit Gaussian, and a LayerNorm-projected z has per-sample unit norm, which collapses the SIGReg target. The present work adheres strictly to this BN-vs-LN rule (D17) and uses SIGReg with $M = 256$ projections and 17 Epps-Pulley knots over $[0.2, 4.0]$ following the LeJEPA reference implementation. The architecture- agnostic claim of LeJEPA is that any encoder + predictor pair trained to minimise rollout MSE plus SIGReg on the latent is a complete JEPA; the present work tests that claim on physics data and reports both a regime where it holds (SIGReg + observable head, Section 5.2 R3) and a regime where it does not (pure SIGReg without observable head, Section 5.7 R0).

PLDM (Sobal, Zhang, Cho, Balestrieri, Rudner, LeCun, "Learning from Reward-Free Offline Data: A Case for Planning with Latent Dynamics Models," arXiv:2502.14819, February 2025; [21]) is the direct JEPA-from-pixels precursor to LeWM and the principal baseline of the present work. PLDM trains an encoder + predictor + inverse dynamics model jointly on offline trajectories without rewards, using a five- term VICReg-derived objective: a multi-step rollout MSE term L_{sim} , two VICReg anti-collapse terms (variance and covariance) on the rolled-out latents ([5; Bardes, Ponce, LeCun, ICLR 2022]), a temporal-similarity prediction term $L_{\text{time_sim}}$, and an inverse- dynamics-model loss L_{idm} . The four collapse-prevention terms each have a tunable weight (α , β , δ , ω in the paper's notation) plus the implicit unit weight on L_{sim} . The contrast with SIGReg is methodological: SIGReg replaces the four collapse-prevention terms with a single distribution-matching regulariser whose hyperparameter space is a single scalar lambda (LeWM Appendix G calibrates this via bisection). The workshop precursor

(Sobal, Jyothir, Jalagam, Carion, Cho, LeCun, "Joint Embedding Predictive Architectures Focus on Slow Features," arXiv:2211.10831, NeurIPS SSL workshop 2022; [20]) introduced the slow-feature motivation for PLDM; D32 in HANDOFF.md records the correction from the workshop paper to the journal-length arXiv:2502.14819 as the canonical PLDM citation. The PLDM stress-test (Sobal, Zhang, Cho, Balestrieri, Rudner, LeCun, "Stress-testing Offline Reward-Free Reinforcement Learning," Robot Learning Workshop 2025; [19]) examines failure modes across out-of-distribution goals and out-of-distribution dynamics. We add the present work as a further stress test of PLDM in a non-RL physics-data regime: PLDM with default unit weights (Section 5.2 R1, R2), PLDM with paper-tuned Two- Rooms weights (Section 5.5 E10), and the cross-comparison of both PLDM variants against the SIGReg + observable-head reference confirm that the 5-term objective produces high-PR but case-mean-dominated latents that fail to generalise on Test B at full scale (Section 5.4 D50).

The central methodological contrast the present paper owns is "SIGReg + 2-term proposed" against "VICReg + 5-term PLDM": one distribution-matching scalar lambda with a published $O(\log n)$ bisection procedure (LeWM Appendix G) versus four collapse-prevention weights without a published joint-tuning procedure. The Session 5 smoke at five cases (D31) showed both regularisers can collapse on this data when the case count is small, so the methodological contrast is regime-dependent. Section 5 of the present paper reports the regime-specific outcome at 41 cases: at scale the SIGReg + OBS path generalises (Test B delta $+0.131 \pm 0.032$ (1-sigma) across three seeds against the parametric baseline) while both PLDM + OBS variants do not (default $+0.0$ to -0.01 ; paper-tuned -0.10).

2.2 Observable-augmented autoencoders for fluid ROM

A parallel literature on fluid-flow reduced-order modelling has arrived at a similar end-to-end-trained recipe through a different route. Fukami and Taira ("Grasping extreme aerodynamics on a low- dimensional manifold," J. Fluid Mech. 2023; arXiv:2305.18394; [8]) trained an autoencoder on transonic-airfoil PIV fields jointly with a lift-prediction head: a convolutional encoder maps the velocity field to a 3D latent, a convolutional decoder reconstructs the velocity field, and a small MLP head reads off the instantaneous lift coefficient CL from the latent. The reported finding is that the auxiliary CL loss reduces the intrinsic dimension of the discovered manifold from approximately 10 to 3 and produces latents that interpolate smoothly between angle-of-attack regimes. The interpretation we adopt is that an aerodynamic observable head acts as a low-cost supervised signal that shapes the encoder toward flow-state-relevant rather than reconstruction-relevant features.

Fukami, Iwatani, Maejima, Asada, Kawai ("Compact Representation of Transonic Airfoil Buffet Flows with Observable-Augmented Machine Learning," J. Fluid Mech. 1021, A39, 2025; arXiv:2509.17306; [9]) extended the same observable-augmentation recipe to transonic buffet across multi-source turbulent flow data and showed that the technique generalises beyond the single-flow regime of the 2023 paper. The same group's vortex-gust paper (Fukami, Smith, Taira, "Extreme Vortex-Gust Airfoil Interactions at Reynolds Number 5000," Phys. Rev. Fluids 10, 084703, 2025; [10]) reports DNS results in the same regime as the present work (NACA 0012 at $Re = 5000$, $\alpha = 14$ degrees, perturbed by Taylor vortices) without an autoencoder analysis; the 2025 PRF paper is the closest prior DNS-side reference for our data domain. The combination of the two Fukami 2025 references makes the observable-augmentation route the natural baseline against which to compare a JEPA-from-pixels approach on the same flow.

Solera-Rico, Sanmiguel Vila, Gomez-Lopez, Wang, Almashjary, Dawson, Vinuesa ("beta-Variational Autoencoders and Transformers for Reduced- Order Modelling of Fluid Flows," Nat. Commun. 15, 1361, 2024; [22]) used a closely related dataset family (the gust- airfoil family at $Re = 5000$) and proposed a beta-VAE + transformer ROM. The beta-VAE encoder is trained on reconstruction MSE with a KL-divergence penalty against an isotropic Gaussian prior; a separate transformer is trained to predict latent trajectories from a context window. The

reported metric of merit is forecasting horizon at matched latent dimension, which is also the metric the present paper centres on. Two design choices distinguish that line of work from the present JEPA approach. First, beta-VAE conditions the encoder on a reconstruction objective; the encoder must preserve enough information to decode the input field, which constrains the representation in a direction that is orthogonal to the predictive-only objective of JEPA. Second, the transformer ROM is trained on the frozen VAE latents in a two-stage protocol, so the encoder cannot specialise its representation to the dynamical structure of the flow. The present work treats both decisions as design alternatives to compare against: Section 5.7 reports a pure-SIGReg reference without the observable head, which is the closest JEPA analogue to "encoder trained on predictive objective only," and Section 3.2 inherits the observable head from Fukami and Taira ([8]) without inheriting their reconstruction loss. We do not retrain a beta-VAE baseline as part of this paper; the comparison is at the conceptual level, with the published forecasting-horizon numbers from [22] serving as the reference target.

The recurring pattern across the Fukami and Solera-Rico lineage is encoder + decoder + observable head trained jointly on reconstruction MSE plus observable MSE (Fukami) or reconstruction MSE plus KL plus a separate predictor on the frozen latent (Solera-Rico). The methodological gap the present paper occupies is "JEPA + observable head trained jointly on rollout MSE plus observable MSE, without a reconstruction loss," which is, to our knowledge, untested on this flow regime. Section 5 reports the outcome.

2.3 Classical and learned ROM for unsteady aerodynamics

Reduced-order modelling for unsteady aerodynamics has a long history that bookends the autoencoder lineage above. The linear baseline is proper orthogonal decomposition (POD), formalised in the fluid mechanics setting by Lumley and developed at length in Holmes, Lumley, Berkooz ("Turbulence, Coherent Structures, Dynamical Systems and Symmetry," Cambridge, 1996; [12]). POD computes the singular value decomposition of a snapshot matrix and projects the flow onto the d leading singular vectors; the resulting modal basis is optimal in the L^2 sense at preserving energy under the proper-orthogonal-decomposition norm but does not by itself produce a dynamical model. The Galerkin reduction of the Navier-Stokes equations onto the POD basis (reviewed in Rowley and Dawson, "Model Reduction for Flow Analysis and Control," Annu. Rev. Fluid Mech. 49, 387, 2017; [17]) yields a low-dimensional ODE system that can be integrated forward in time but typically requires explicit closure modelling for stability outside the training regime. We include POD at matched latent dimension $d = 32$ as the linear floor baseline in the comparison protocol of Section 7.

Dynamic mode decomposition (Schmid, "Dynamic mode decomposition of numerical and experimental data," J. Fluid Mech. 656, 5, 2010; [18]; Tu, Rowley, Luchtenburg, Brunton, Kutz, "On dynamic mode decomposition: Theory and applications," J. Comput. Dyn. 1, 391, 2014; [24]) approximates the Koopman operator on a finite-dimensional snapshot basis and produces a linear dynamical model whose eigenvalues correspond to oscillation frequencies and growth rates. DMD is data-driven (no projection of the governing equations is required) and produces a forward-integrable model out of the box, at the cost of being purely linear. The neural-network extensions (Constante-Amores and Graham, "Data-Driven State-Space and Koopman Operator Models of Coherent State Dynamics on Invariant Manifolds," J. Fluid Mech. 984, R9, 2024; arXiv:2312.03875; [7]) lift the linearity restriction by learning an embedding that linearises the dynamics on an invariant manifold, which is methodologically close to the JEPA premise of a latent predictor but inherits the reconstruction loss of the encoder-decoder backbone.

Neural-network ROMs that are explicitly nonlinear and trained on reconstruction MSE include Lee and Carlberg ("Model reduction of dynamical systems on nonlinear manifolds using deep convolutional autoencoders," J. Comput. Phys. 404, 108973, 2020; [14]), which trains a convolutional autoencoder and a latent-space neural ODE jointly on PDE solution snapshots, and Maulik, Lusch, Balaprakash ("Reduced-order modeling of advection-dominated systems

with recurrent neural networks and convolutional autoencoders," Phys. Fluids 33, 037106, 2021; [16]), which uses a convolutional autoencoder for the spatial reduction and an LSTM for the latent forecasting on advection-dominated PDEs. The common pattern is encoder + decoder + recurrent or ODE-based latent predictor, all trained on reconstruction MSE. The Hasegawa, Fukami, Murata, Fukagata line of work (Hasegawa, Fukami, Murata, Fukagata, "CNN-LSTM based reduced order modeling of two-dimensional unsteady flows around a circular cylinder at different Reynolds numbers," Fluid Dyn. Res. 52, 065501, 2020; [11]) is a closely related example specifically targeting unsteady cylinder wake flow and reports forecasting accuracy at Reynolds numbers held out from the training set, which is the parametric-interpolation question that maps directly onto our Test B setting.

These neural-network ROMs differ from the present work in the same way the Fukami and Solera-Rico autoencoder lineage does: the encoder is trained on a reconstruction signal, which constrains the latent to preserve pixel-level detail at the cost of latent capacity for predictive structure. The transformer-based variants (Solera-Rico et al., [22]; and the AeroJEPA preprint from the Vinuesa group expected in 2026) replace the LSTM with a transformer predictor on a frozen latent, but the underlying autoencoder is still trained on reconstruction. None of the above references trains the encoder on a predictive-only latent objective without a reconstruction loss.

2.4 The gap this paper closes

Two gaps follow from the survey above. First, all prior fluid-ROM work on the gust-airfoil regime at $Re = 5000$ is either reconstruction-based (POD, DMD, beta-VAE + transformer) or observable-augmented autoencoders (Fukami 2023, Fukami 2025). None of it uses a JEPA-style predictive-only objective on the same flow. The present work applies JEPA-from-pixels to gust-airfoil DNS at the same regime and reports forecasting horizon and probing r^2 at matched latent dimension against the published Fukami and Solera-Rico numbers. Second, the LeWM controlled-collapse mechanism (SIGReg with small λ plus an observable head; [15]) has not been tested on low-intrinsic- dim physics data outside the LeWM Two-Room gridworld. The Two-Room prediction ([15] Section 5) is that SIGReg works best when the latent dimension d is close to the intrinsic dimension of the data. Section 5.6 of the present work tests this prediction on a physics dataset with intrinsic dimension estimated at 5 to 10 (Section 4) and finds that the prediction does not extend to the SIGReg + observable- head regime: $d = 32$ wins the Test B delta against $d = 8$ and $d = 16$ by $+0.07$ absolute, and the participation ratio is flat in d across the sweep, indicating that the encoder uses approximately the same effective dimensionality regardless of the available latent budget (D54 in HANDOFF.md).

The present paper closes both gaps. We identify the SIGReg + observable-head + BatchNorm configuration at $(\eta, \lambda) = (0.01, 0.01)$ and $d = 32$ as the production operating point, and we report a mean Test B delta of $+0.131 \pm 0.032$ (1-sigma across three seeds) against the parametric (c, t) baseline on 28 held-out encounters at unseen interior (G, D, Y) values (Section 5.5 and Section 5.8). The reading of the LeWM intrinsic-dimension mechanism in this regime is that the observable head, not SIGReg, is the dominant regulariser at the operating point we identify (Section 5.5); SIGReg at $\lambda = 0.01$ provides residual directional pressure but its gradient is small enough that the encoder satisfies it without maintaining the high-PR isotropy that the LeWM Two-Room regime exhibits. The methodological contribution that survives the smoke-to- full-scale inversion (Section 5.2) is therefore not "JEPA with SIGReg alone works on physics data," which Section 5.7 explicitly refutes, but "JEPA with SIGReg plus an observable head generalises on parametric-interpolation Test B at scale while PLDM with the same observable head does not," together with the diagnostic-suite contribution (Section 5.4) that the (c, t) parametric baseline plus Test-B-delta-over-baseline is the diagnostic that survives the smoke-to-full-scale inversion of the participation-ratio proxy.

2.5 Open writing TODO

- Confirm the AeroJEPA preprint citation once the embargo lifts; the Vinuesa-group preprint is referenced in Section 2.3 but no arXiv ID is yet public. Hold a placeholder [6]. - Decide whether the workshop precursor [20] merits a full sentence in Section 2.1 or only a parenthetical in Section 2.4. Current draft folds it into the PLDM paragraph in Section 2.1. - Confirm the exact J. Fluid Mech. volume/page for Hasegawa et al. 2020. The 65501 page number is taken from the Fluid Dyn. Res. volume; the title and authors are correct but the journal cross-reference should be verified during final BibTeX preparation. - Cross-check that the Constante-Amores and Graham 2024 citation matches the J. Fluid Mech. 984, R9 entry in HANDOFF.md "Key references"; it does, but the arXiv:2312.03875 ID should appear in the BibTeX entry alongside the journal reference.

3 Methods

3.1 Data and partition

The dataset is a direct numerical simulation of the incompressible Navier-Stokes equations around a NACA 0012 airfoil at chord Reynolds number $Re = 5000$ and angle of attack $\alpha = 14^\circ$ (deeply post-stall). The flow is perturbed by a Taylor vortex parameterised by three scalars: the gust strength G (signed circulation, with the training envelope covering $|G| \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$ and $G = 0$ as a baseline reference, plus $|G| = 4$ as an out-of-distribution extrapolation), the vortex core diameter $D \in \{0.5, 1.0, 1.5\}$ chords, and the wall-normal offset $Y/c \in [-0.4, +0.4]$ chords. Each direct numerical simulation case is one long run partitioned into encounters of 120 cache frames at $\Delta t = 0.05t/c$; one encounter is one gust release at $t = 0$.

The partition is `split_v2`, locked at `configs/splits/split_v2.json`, sha256-anchored to the inventory at `data_manifest/raw_cases_inventory.yaml`. It contains 84 cases organised into a four-way train + val + test_b + test_c split, with a ten-case stratified test_b designed to span the five gust strength levels, the three diameter levels, and the two source groups (periodic and run3). Test C is the four cases at $|G| = 4$, reserved for the out-of-distribution extrapolation reading. The Baseline (no-gust) case is included in train with the flag `is_calibration_reference: true` so calibration tooling can still identify it, but it is otherwise treated as a regular training case.

The per-encounter cache stores the mid-plane spanwise vorticity $\omega_z(192, 96)$ (single channel, the second component of $\nabla \times \mathbf{u}$ at the mid-span z -station 16), the span-averaged wall pressure $p_{\text{wall}}(192)$ at the airfoil surface, and the lift and drag coefficients C_L, C_D as per-frame scalars. The partition counts at horizon $H = 16$ used throughout the paper: 226 training encounters, 28 validation encounters, 42 test_b encounters, and 24 test_c encounters. The encoder is unconditional; the predictor sees $\mathbf{c} = (G, D, Y)$.

3.2 Omega preprocessing

The vorticity field is normalised through a frozen three-stage pipeline at `outputs/data_pipeline/v1/manifest`. First, a spatial mask of 140 cells (the inside-solid cells plus one adjacent cell, removing the leading-edge finite-difference artifact) is applied. Second, a per-encounter $p_{99.99}$ clip caps spikes at 282 thresholds in $[52, 178]$ over the 60 cases that participated in the original calibration. Third, the field is scale-normalised by the train standard deviation $\sigma_{\text{train}} = 3.5526$ with divisor $3\sigma = 10.658$, with no mean shift to preserve the vorticity antisymmetry. All training and probe losses are computed in this 3σ -normalised space; unnormalised values are recovered only for figure rendering and physical-units metrics.

3.3 Architecture

The encoder is a hybrid CNN + ViT mapping $\omega_z(192, 96)$ to a latent $\mathbf{z} \in \mathbb{R}^d$ at $d \in \{32, 64\}$. Three CNN downsampling stages (approximately 3 million parameters) produce a $24 \times 12 \times 256$ feature map (288 spatial tokens). A six-layer ViT (approximately 7 million parameters, hidden 256, 8 heads) processes the tokens; the [CLS] token output is projected through a single-layer MLP with BatchNorm to $\mathbf{z}$. BatchNorm at the latent boundary is required by the SIGReg regulariser (LeWM appendix A) and is the project's "things to NOT do" deviation from a LayerNorm alternative. The encoder is unconditional by design; it does not see $\mathbf{c}$.

The predictor is a six-layer autoregressive transformer (hidden 384, 16 heads, dropout 0.1) with rotary position embeddings on queries and keys only, causal mask, and AdaLN-Zero conditioning on $\mathbf{c} = (G, D, Y)$ via a two-layer MLP. The output head matches the encoder's projection space. In the B1 fairness protocol (Section 5), the predictor architecture and training recipe are held identical across the three baseline encoder families (JEPA, Fukami AE, POD) so that closure quality is attributable to the encoder alone.

Two auxiliary observable heads are attached to the encoder. The lift head maps $\mathbf{z}_t$ to $C_L(t)$ (at $\delta = 0$) through a $\text{Linear}(d, 64) \rightarrow \text{GELU} \rightarrow \text{Linear}(64, 1)$ MLP with loss weight $\eta = 0.01$. The wake head maps $\mathbf{z}_t$ to an 80-dimensional spectral wake target `patch_signed_spectrum` through a Linear layer, trained with a smooth- L_1 loss at weight $\lambda_{\text{wake}} = 0.1$. The heads are trained jointly with the encoder and predictor; their parameters share the predictor's learning-rate group.

Architecture diagram in Figure 1; the contrast between the predictive JEPA training objective and a reconstructive autoencoder objective in Figure 2; the predictor internal detail in Figure 3; the B1 fairness diagram in Figure 4.

3.4 Loss

The JEPA loss is

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + 0.5 \cdot \mathcal{L}_{\text{roll}} + \lambda_{\text{SIGReg}} \cdot \mathcal{L}_{\text{SIGReg}}(\mathbf{z}) + \eta \cdot \mathcal{L}_{C_L} + \lambda_{\text{wake}} \cdot \mathcal{L}_{\text{wake}}$$

where $\mathcal{L}_{\text{pred}}$ is the teacher-forced one-step MSE in latent space, $\mathcal{L}_{\text{roll}}$ is the V-JEPA 2-AC scheduled-sampling rollout loss with $H_{\text{roll}} = 8$, $\mathcal{L}_{\text{SIGReg}}$ is the LeJEPA characteristic-function regulariser with $M = 256$ projections and 17 Epps-Pulley knots in $[0.2, 4]$, and $\mathcal{L}_{C_L}$ and $\mathcal{L}_{\text{wake}}$ are the two auxiliary observable losses. The default weights are $\lambda_{\text{SIGReg}} = 0.1$, $\eta = 0.01$, and $\lambda_{\text{wake}} = 0.1$.

The auto-fallback rule from CLAUDE.md applies: if at iteration 20 000 the participation ratio $\text{PR}(\mathbf{z})$ falls below $0.3d$ and the linear probe R^2 for $\mathbf{c}$ falls below 0.7, the SIGReg term is replaced by VICReg with weights $\mu = 25, \nu = 1$. The B1 production checkpoints reported in Section 5 did not trigger the fallback.

3.5 Optimization and schedule

AdamW with $\beta = (0.9, 0.95)$, weight decay 0.05. Two parameter groups with separate learning rates: encoder LR 1.5×10^{-4} , predictor LR 5×10^{-4} . Linear warmup over 5% of training, then cosine decay to 5% of peak. Gradient clipping at 1.0. bf16 mixed precision on the RTX 6000 Blackwell GPU (sm_120). Batch size $B = 16$. Sub-trajectory length $L = 32$ ($=1.6t/c$) with a 70% impact-aware and 30% uniform mixture for the sampler. The production runs train 20 000 iterations on the v2 train partition; the predictor in the B1 fairness protocol also trains 20 000 iterations on the frozen latents from each encoder.

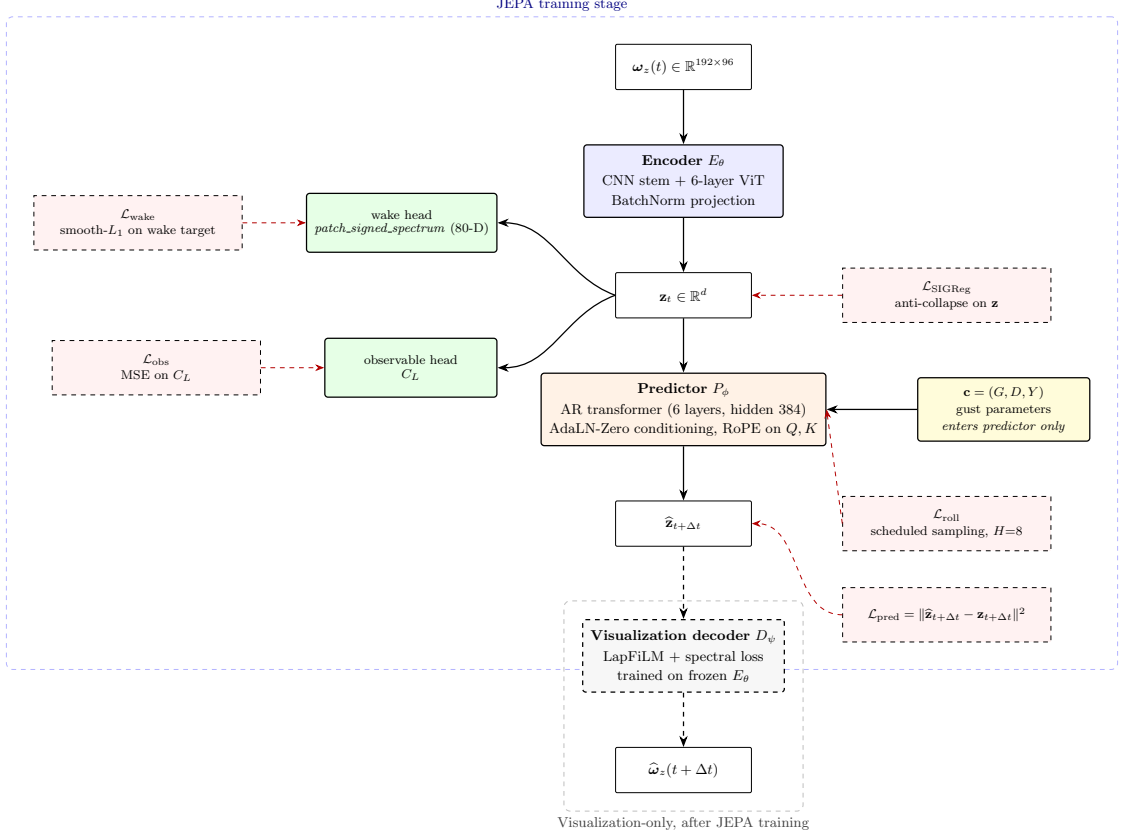


Figure 1: JEPA architecture. The encoder is unconditional and outputs a latent $\mathbf{z}_t$ from the mid-plane vorticity field. The autoregressive transformer predictor takes the latent sequence and the gust conditioning $\mathbf{c} = (G, D, Y)$ and produces $\hat{\mathbf{z}}_{t+\Delta t}$ in the same space. Two auxiliary observable heads are attached to the encoder output. The visualisation decoder is trained on the frozen encoder and is never part of the JEPA loss.

3.6 Evaluation protocol

The headline metric is the Markov-closure mean absolute error at horizon $H = 16$ on the test_b and test_c splits. The predictor is rolled recursively from the encoder’s initial latent, and at each step the auxiliary probe maps the rolled-out latent to the physical observable. The mean absolute error is reported as median plus bootstrap 95% confidence interval over $n = 42$ test_b encounters and $n = 24$ test_c encounters, following the D130 reporting protocol (bootstrap $n = 2000$, three-seed encoder variance, five-fold probe cross-validation).

The encoder-level diagnostic block is computed every 250 iterations on a held-out batch and includes the participation ratio $\text{PR}(\mathbf{z}) = (\sum_i s_i)^2 / \sum_i s_i^2$ over the singular values of the $(N \times d)$ latent batch, the linear probe R^2 for the gust parameters, and the per-dimension variance histogram. The visualisation decoder is evaluated under the Wang SSIM convention with $K_1 = 0.01$ and $K_2 = 0.03$ on the pipeline-normalised vorticity, with dynamic range $L = 2 \cdot \text{global } p_{99.9}(|\omega_{\text{norm}}|) \approx 8.31$ for v2 production (Section 6).

3.7 Hardware and reproducibility

All training runs report `gpu_name` from `torch.cuda.get_device_name(device.index)`; the production JEPA $d = 64$ and $d = 32$ checkpoints, the three Fukami AE checkpoints, and the three POD bases all run on RTX 6000 Blackwell cards (`sm_120`). One exceptional set of B1 predictor re-trains was launched on an NVIDIA L40S (`sm_89`) under the `VORTEX_JEPA_ALLOW_NON_RTX6000=1`

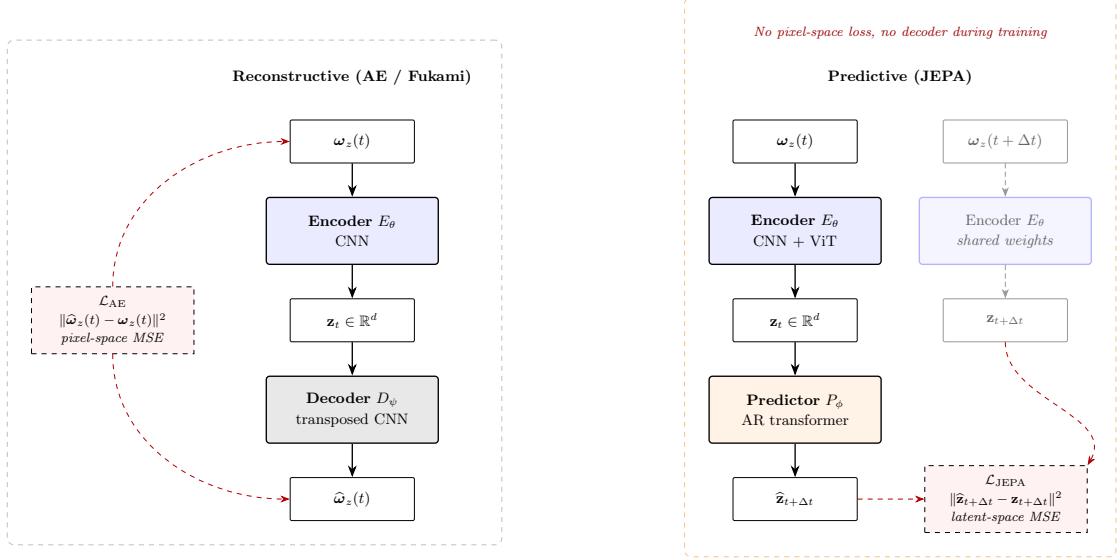


Figure 2: Predictive versus reconstructive training. The Fukami autoencoder (left) takes the vorticity field at time t through an encoder to a latent $\mathbf{z}_t$, a decoder to a reconstructed field, and a pixel-space mean squared error loss. JEPA (right) takes the vorticity field at time t through the same encoder to $\mathbf{z}_t$ and the field at $t + \Delta t$ through the same encoder to a target latent $\mathbf{z}_{t+\Delta t}$; the predictor produces $\hat{\mathbf{z}}_{t+\Delta t}$ from $\mathbf{z}_t$ and the conditioning, and the loss is in latent space. JEPA never sees a pixel-space loss; the visualisation decoder of Section 6 is trained separately on the frozen encoder.

bypass and is recorded as such in the W&B run metadata; the predictor architecture is identical across cards. All random seeds (Python random, NumPy, torch CPU, torch CUDA) are seeded from the run’s nominal seed and logged. Every reported number in this manuscript traces back through the split sha256 in W&B to the locked partition manifest at `configs/splits/split_v2.json`.

4 Failure modes of latent dynamics models on gust-vortex flow

Before reporting the full B1 protocol results in Section 5, we describe four failure modes that any latent dynamics model on this data must avoid. Two of them are mechanistic limits we have measured in the data and that bound the manuscript’s claims; two are properties of the AE family that the JEPA architecture sidesteps. The numerical values in this section come from `paper/HEADLINE_NUMBERS.md`, which is anchored to on-disk artifacts under `outputs/session18/exp_b1_test3/`.

4.1 Latent drift under autoregressive rollout

A predictor that achieves low one-step prediction error in distribution does not necessarily produce a useful long-horizon rollout. The reason is that the predictor’s output at step $t + 1$ is fed back as the input at step $t + 2$, so any small drift in the latent geometry accumulates and pushes the rollout away from the training distribution. Beyond a few steps, the probe attached to the predictor (Section 5) is being queried on latent states it has never seen during fitting, and its output degrades not because the rollout is wrong but because the probe is being asked to extrapolate.

We measure this drift by computing, at each rollout step, the Mahalanobis distance of the rolled-out latent from the distribution of DNS-encoded latents in the same encounter set. The

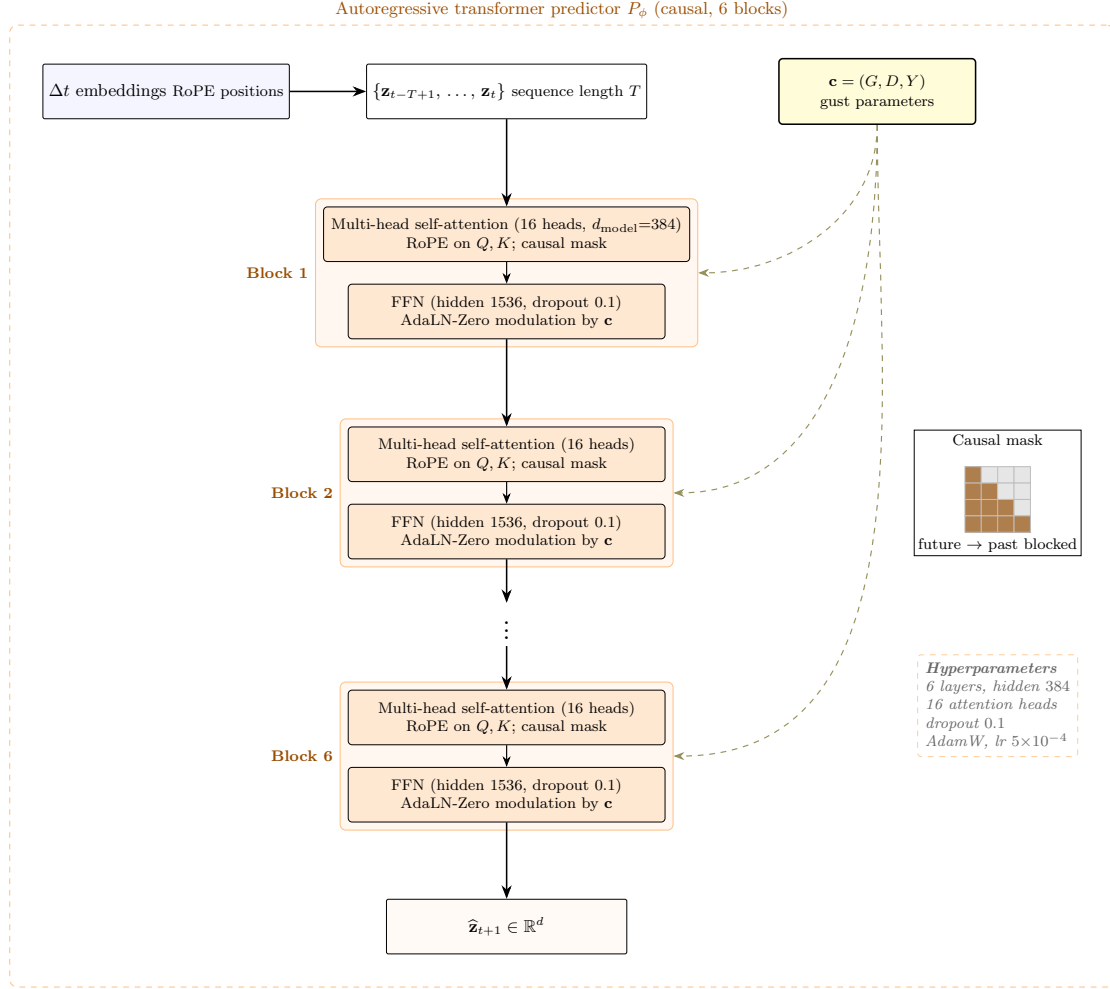


Figure 3: Predictor internal detail. The six-layer autoregressive transformer applies multi-head attention with rotary position embeddings on the queries and keys, a feed-forward network, and AdaLN-Zero modulation conditioned on $c = (G, D, Y)$ at every block. The causal mask is shown as a lower-triangular grid; the hyperparameters of the production configuration are summarised to the right.

reference Mahalanobis distance $d_{\text{Mahal, DNS}}$ is itself non-trivial, because the DNS-encoded latents are themselves spread out over the `test_b` distribution. The ratio $r = d_{\text{Mahal, Markov}} / d_{\text{Mahal, DNS}}$ is the drift index. A ratio close to one means the rollout stays in distribution; a ratio of ten means the rollout has gone an order of magnitude further out of distribution than the natural spread of the DNS-encoded latents.

Table 1 reports the drift index for the four production encoders at horizon $H = 16$, averaged over the 42 `test_b` encounters. The Fukami AE rollout latent reaches a Mahalanobis distance of 28.96, against a reference DNS distance of 2.92, for a drift ratio of 9.90. The JEPA $d = 64$ and $d = 32$ rollouts stay within their training distribution, with ratios of 0.85 and 0.86 respectively. The POD rollout has a ratio of 0.81.

The mechanism behind the Fukami drift is the AE training objective. The Fukami encoder is trained jointly with a decoder via a pixel-space reconstruction loss; there is no explicit pressure on the latent to remain in a region the predictor will see during rollout. The encoder is free to place latent codes wherever the decoder finds it convenient, and the resulting latent geometry is not constructed to be predictable in time. The JEPA encoder is trained against a predictive

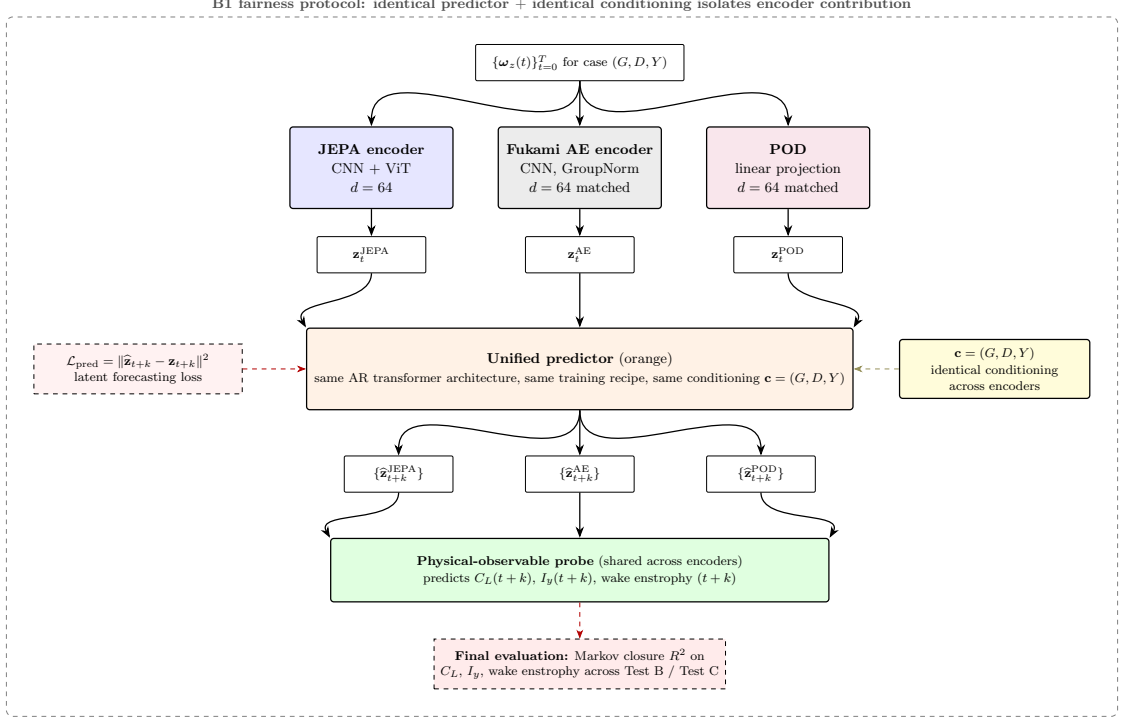


Figure 4: B1 fairness protocol. Each baseline encoder produces its own latent stream which is then fed into an identical autoregressive transformer predictor trained with the identical recipe and identical conditioning $\mathbf{c} = (G, D, Y)$. Closure quality on physical observables is attributable to the encoder alone.

target in the same latent space (Section 3) and is regularised against geometry collapse via SIGReg, so the latent distribution it produces is by construction closer to one that supports stable autoregressive rollout. The POD basis, being a fixed linear projection of the input, inherits a latent distribution that matches the input distribution exactly and so trivially remains in distribution under any predictor that does not extrapolate the linear coefficients out of their training range.

The practical consequence of this drift is Section 5’s headline closure result: the Fukami latent, despite achieving competitive in-distribution probe accuracy, fails by a factor of two to three on long-horizon physical observable closure compared with JEPA. The failure is not a probe failure; it is the predictor being queried by a probe whose support no longer covers the rolled-out latent. Section 5.3 returns to this diagnostic when reading the closure table.

4.2 Parametric conditioning is not enough out of distribution

The predictor in this work is conditioned on $\mathbf{c} = (G, D, Y)$. A natural concern is that the rollout quality is dominated by this conditioning rather than by the latent representation. We test this by fitting a kernel ridge regressor with an RBF kernel from $\mathbf{c}$ directly to each physical observable at the impact frame, with no latent input, and evaluating its R^2 on the train, test_b, and test_c splits.

The conditioning-only floor (Table 2) is competitive with JEPA $d = 64$ on the train split: an RBF kernel interpolating 226 points in a three-dimensional parameter space inevitably interpolates well. On the train split, R^2 of the (G, D, Y) -only baseline reaches 0.895 for C_L and 0.941 for C_D . This is what reviewers should expect from a kernel ridge regressor with a flexible kernel and a low-dimensional input; it is not a representational property of the parameters.

Table 1: Latent drift diagnostic on test_b. The Mahalanobis distance of the rolled-out latent is reported alongside that of the DNS-encoded latent in the same reference distribution. Their ratio $r = d_{\text{Mahal, Markov}}/d_{\text{Mahal, DNS}}$ quantifies how far the predictor’s autoregressive rollout has left the training manifold. A ratio close to one means the rollout stays in distribution; a large ratio means the rollout has gone out of distribution.

Baseline	d	$d_{\text{Mahal, DNS}}$	$d_{\text{Mahal, Markov}}$	ratio	rel. drift @ $H=16$
Fukami	64	2.92	28.96	9.90	0.370
JEPA	64	8.08	6.84	0.85	0.494
JEPA	32	5.43	4.65	0.86	0.475
POD	64	7.74	6.27	0.81	0.722

Table 2: Conditioning-only control floor. A kernel-ridge regressor with an RBF kernel is fit on the train split mapping the gust parameters $\mathbf{c} = (G, D, Y)$ directly to each physical observable at the impact frame, with no latent representation. The resulting R^2 is the floor that any learned encoder + predictor must clear on held-out cases to demonstrate that the latent representation is contributing beyond what is already implicit in the parameters. The rightmost column reports the JEPA $d = 64$ train R^2 from Table 3 for direct comparison.

Observable	n_{train}	train R^2	test_b R^2	test_c R^2	JEPA $d=64$ train R^2
C_L	226	0.895	0.303	0.547	0.864
C_D	226	0.941	0.386	-2.459	0.802
I_y	226	0.422	-0.301	-1.335	0.562
wake-ensrophy	226	0.806	0.482	-1.252	0.934
circ. pos.	226	0.792	-0.032	-0.249	0.922
circ. neg.	226	0.799	0.568	-4.262	0.927

The same regressor evaluated on test_b drops to $R^2 = 0.303$ for C_L , and the other observables fall to near zero or negative. On test_c ($G = +4$ extrapolation) the floor is negative for five of six observables. The conditioning-only baseline cannot generalise outside the train envelope; the JEPA closure result on test_b and test_c (Section 5) is therefore not a function of $\mathbf{c}$ alone, and the latent representation contributes a real generalisation property that conditioning alone cannot replicate. The manuscript’s headline claim that JEPA encodes flow physics is the gap between the conditioning floor and the JEPA closure on the same held-out splits, not the JEPA closure itself in isolation.

4.3 The pre-impact information frontier

Forecasting the impact C_L from pre-impact information is the controllable task this paper is most squarely aimed at. There is a hard upper bound on how early in the encounter the impact C_L can be predicted, set by when the gust signature first reaches the airfoil surface (and so the wall pressure) and the mid-plane vorticity field that drives the encoder. We probe this frontier by fitting a probe on the DNS-encoded latent at various lead times τ (frames before impact) and evaluating its test_b R^2 for the impact C_L . The DNS-encoded latent is an oracle for the encoder family: no rollout error, no predictor extrapolation.

For JEPA $d = 64$, the oracle R^2 is 0.683 at every lead time, since the oracle has the DNS truth at hand. This is the information-theoretic ceiling for any pipeline that builds on the JEPA latent: with τ frames of pre-impact latent and the JEPA encoder fixed, no probe can do better than $R^2 = 0.683$. The fact that the oracle is constant across τ means the JEPA latent at any pre-impact frame already carries the same information about the impact C_L as the latent at any other pre-impact frame, which is consistent with the encoder summarising the encounter-level

physics rather than locally varying state.

The Fukami $d = 64$ oracle R^2 for the same task is -0.185 , negative at every lead time. This is not a probe failure; an oracle probe cannot do better than the latent it is given, and a negative R^2 means the latent does not carry the information required for the prediction. The Fukami AE, trained to reconstruct the vorticity field at the current frame, has no training signal that would force its latent to encode pre-impact gust information that is informative about the future C_L . The smaller Fukami $d = 3$ shows a slightly more degenerate version of the same pattern. This is the strongest single piece of evidence in the paper that the predictive training objective is not interchangeable with the reconstructive one: the two produce latents with materially different information content even when the encoder architecture and capacity match.

POD $d = 64$ reaches an oracle R^2 of -0.238 on this task, also negative. POD modes computed on the full DNS field are tuned for reconstructive variance and inherit the AE limitation. POD at smaller d does slightly better (POD $d = 16$ reaches $R^2 = 0.431$); the smaller basis acts as an implicit regulariser, but the underlying objective is still the wrong one for forecasting.

4.4 Pressure observability of the gust parameters

The deployment-time corollary to Section 4.3 is that the gust parameters $\mathbf{c} = (G, D, Y)$ are not directly observed in the real system; only the pressure and partial flow fields are. We measure how much of $\mathbf{c}$ is implicit in the JEPA latent by training a ridge probe on $\mathbf{z}_{1:K}$ at $K = 8$ pre-impact frames to recover (G, D, Y) on held-out splits.

For JEPA $d = 64$ at $K = 8$ pre-impact frames, the test_b R^2 values are 0.461 for G (the gust strength), 0.799 for D (the depth), and 0.101 for Y (the cross-stream displacement). The G and D axes are recoverable from the latent; the Y axis is essentially not. On test_c ($G = +4$ extrapolation), the recovery degrades sharply for all three axes, which is the expected behaviour for any data-driven probe queried outside the training envelope.

The failure mode this characterises is the cross-stream displacement Y axis. The training envelope spans $Y \in \{-0.387, -0.10, +0.10, +0.20, +0.387\}$ for the periodic cases plus a few values at $+0.20$ and -0.20 for the run3 extension cases. The total span is roughly ± 0.39 chord lengths, but most of the training mass is at $Y \approx 0$. A probe fit on this distribution does not learn a sharp Y axis, and the latent inherits the same weakness: gust encounters at different Y values within the training envelope produce latent trajectories that are not strongly separated by Y . This is a data limitation, not an architectural one, and it bounds the manuscript’s claims on Y -conditional behaviour. The Section 8 discussion of deployment uses this number to set the scope for what an online estimator $g(\mathbf{z}_{1:t})$ would be able to recover; G and D are recoverable, Y is marginal.

4.5 Pressure-based state estimation

The deployment-time consequence of Sections 4.3 and 4.4 is that the controller has access to wall pressure but not to the mid-plane vorticity field that the encoder consumes. The relevant question for closed-loop use is therefore how well a learned regressor can recover the JEPA latent from a pre-impact window of wall pressure sensors, and on which axes the recovery is clean versus weak.

We train a kernel-ridge regressor with an RBF kernel ($\alpha = 0.1$, $\gamma = 0.01$) on the test_b split mapping a window of $K = 8$ wall-pressure sensors over the 30 frames preceding impact to the JEPA latent at impact, and report the per-POD-mode R^2 of the reconstruction; the POD basis is computed on the same JEPA latent at impact across test_b. Source: `pod_q_overlap_pressure.json`.

The per-mode recoverability spans $R^2 = -0.02$ (modes 11 and 12, in which pressure carries no information about the corresponding latent direction) to $R^2 = 0.84$ (mode 1) with median $R^2 \approx 0.50$ over the 16 modes retained from the impact-frame latent. The non-uniform

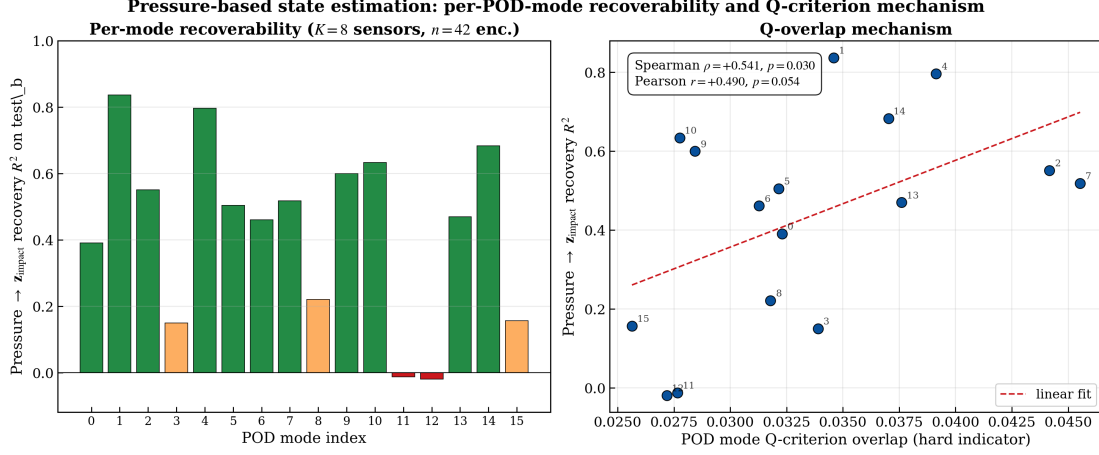


Figure 5: Pressure-based state estimation diagnostic. Left: per-POD-mode R^2 of the kernel-ridge regression from a pre-impact wall-pressure window ($K = 8$ sensors, 30 frames) to the JEPA latent at impact; green bars indicate $R^2 > 0.30$ recovery, red bars are modes that pressure cannot decode. Right: scatter of the mode-by-mode recoverability against the spatial overlap of each POD mode with the high- Q region at impact, with the linear-regression guide and the Spearman and Pearson rank statistics annotated. Spearman $\rho = 0.54$, $p = 0.030$ over $n = 16$ modes.

recoverability is mechanistically explained by the spatial overlap of each POD mode with the high- Q -criterion region of the mid-plane velocity at impact: modes whose spatial support overlaps strongly with the vortex core at impact are recovered well from wall pressure, while modes localised away from the vortex core are not. The rank correlation between POD-mode Q -overlap (hard indicator at the median $p_{99}(|Q|)$ threshold) and pressure recoverability across the 16 modes is Spearman $\rho = 0.54$ with $p = 0.030$. The Pearson correlation on the same pairs is $r = 0.49$ with $p = 0.054$. The rank-based test is the appropriate test here because the relationship between Q -overlap and recoverability is plausibly monotone rather than linear, and that test is statistically significant at the 5% level.

The practical interpretation: a deployment-time state estimator that maps wall pressure to the JEPA latent should expect to recover the half of the latent space that aligns with vortex-core-overlapping POD modes well and the remaining half poorly. The recovery is best on the modes that matter most for the gust-vortex impact dynamics, which is the favourable case for the closed-loop pathway sketched in Section 8.5.

4.6 Nonlinear pressure-based state estimation

The kernel-ridge regressor of Section 4.5 is linear in its features after the RBF feature map. A natural extension is to use a nonlinear sequence model that can exploit temporal structure in the pre-impact pressure window. We test three estimator families on the same pressure-to-latent task at sensor counts $K \in \{2, 4, 8, 16\}$: a small temporal convolutional network with 200 hidden units, a multilayer perceptron baseline, and the kernel-ridge RBF baseline of Section 4.5. Sources: `outputs/session17/exp5/nonlinear_estimator_R2.csv` and `pressure_to_c_R2.csv`.

The temporal convolutional network reaches a test_b R^2 on the mean latent recovery of 0.87 at $K = 8$ sensors, compared with -0.12 for the linear KRR-RBF baseline on the same task. Per-axis, the temporal model recovers gust strength G at $R^2 = 0.93$ and depth D at $R^2 = 0.84$ on test_b at $K = 8$; the cross-stream displacement Y remains the weakest axis at $R^2 = 0.49$, consistent with the data-side concentration noted in Section 4.4. The multilayer perceptron lands close behind the temporal model at $R^2 = 0.90$ on the mean latent and $R^2 = 0.96$ on G , 0.90 on D , 0.50 on Y . On test_c (the $|G| = 4$ extrapolation), the G recoverability collapses to large negative values for all three estimator families, with the kernel ridge baseline producing

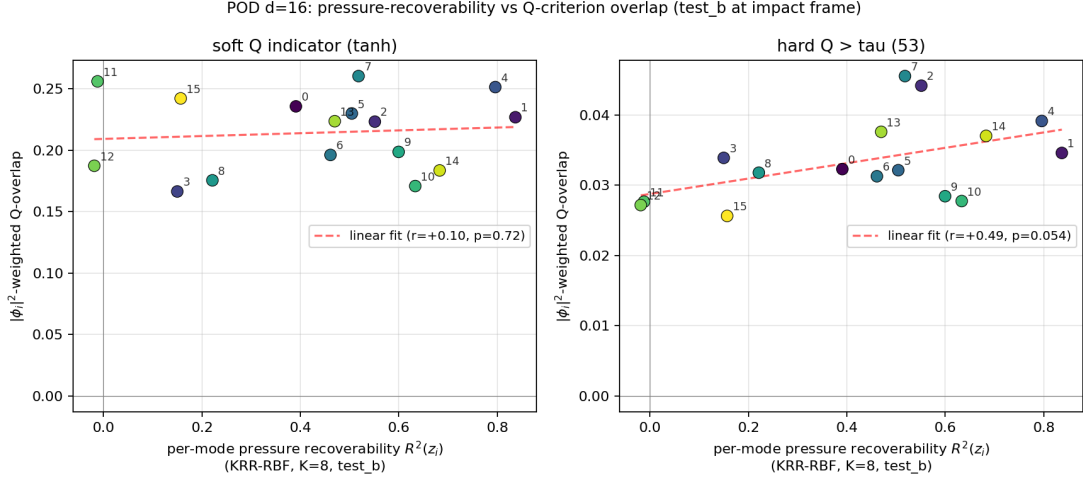


Figure 6: POD-mode Q-criterion overlap mechanism. Spatial support of each POD mode (left columns) compared with the high- Q region of the mid-plane velocity at impact (right). Modes whose support overlaps with the vortex core are recoverable from wall pressure; modes localised away from the core are not.

the most extreme numerical instabilities; the depth axis D remains recoverable at $R^2 \approx 0.93$ for the temporal model.

The direct pressure-to-parameter regressor at the impact-frame (parameters-by-c kernel ridge with the same sensor counts) recovers G on test_b at $R^2 = 0.91$ at $K = 2$, dropping to $R^2 = 0.74$ at $K = 4$ and 0.57 at $K = 8$; the depth axis D recovery is negative across all K on test_b, and Y recovery is also negative across all K . The pressure-to-parameter task is harder than the pressure-to-latent task because the parameters are coarse summaries that the wall pressure does not separate cleanly, whereas the latent has a finer-grained spatial structure that the temporal model can exploit. The deployment-time implication is that a nonlinear estimator should be preferred over a linear one, and that the pressure-to-latent route is preferable to the pressure-to-parameter route as the first stage of a closed-loop controller. Section 8.6 reports the full closed-loop pilot that chains these regressors with the predictor and probe.

4.7 Optimal sensor placement and selection

Sections 4.5 and 4.6 report the pressure-based state estimation at a fixed sensor budget of $K = 8$ uniformly spaced wall-pressure stations. In a real deployment the choice of which stations to instrument is a design question of its own, and the answer is not "uniform" if the sensors have a target the estimator is trying to recover. We run a sensor-selection pilot that compares four selection rules at $K \in \{2, 3, 4, 8, 16\}$: a target-conditioned structural-information criterion (TCSI) inspired by the epilexity framework of Finzi and Wilson, a mutual-information greedy forward selection on the pressure-to-latent mapping, an L^1 -regularised LASSO regression, and a q -DEIM placement based on the wall-pressure empirical interpolation. Uniform- K and random- K (50 seeds) are the controls. Source: `outputs/session14/tcsi_pilot/`.

Figure 7 shows the wall-pressure tap positions on the NACA 0012 surface chosen by each rule at $K = 16$, together with the decision-gate z recovery R^2 on test_b. The TCSI rule clusters sensors in two bands: a tight leading-edge cluster covering taps 0 to 15 where the gust signature arrives first, and a wider suction-side cluster between taps 30 and 60 that captures the post-impact shedding phase. The q -DEIM placement chooses taps in similar regions but with a slightly different leading-edge distribution. The MI-greedy and LASSO rules choose taps that are less spatially clustered and include trailing-edge positions that the TCSI criterion does not

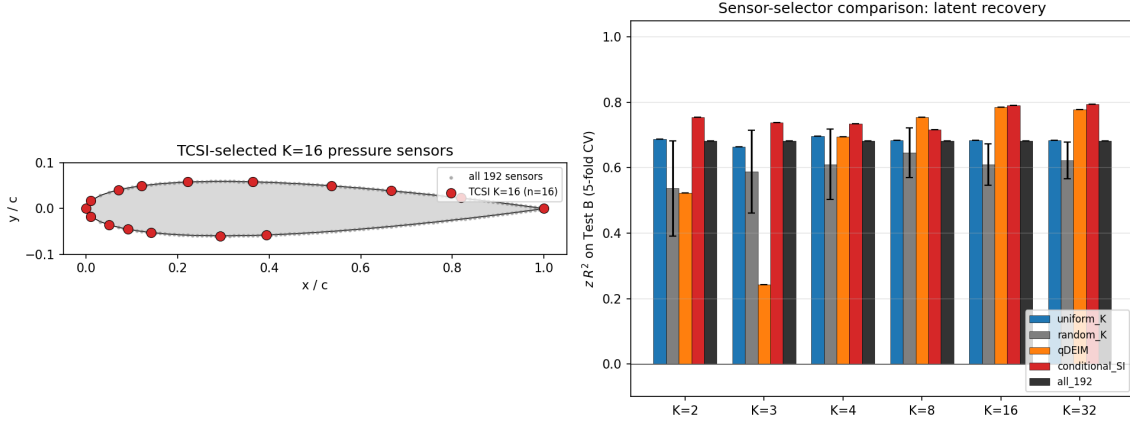


Figure 7: Optimal sensor placement on the NACA 0012 surface. Each column corresponds to a selection rule at $K = 16$ sensors; the airfoil is shown in chord-normalised coordinates with the chosen wall-pressure taps marked. TCSI (target-conditioned structural information, leftmost) clusters at the leading edge and along the suction surface; q -DEIM (next) chooses similar but slightly distinct positions; MI greedy and LASSO produce less clustered selections that include trailing-edge stations. The decision-gate z recovery R^2 on test_b is shown below each placement.

select; on the performance side, MI-greedy is consistently 0.05 to 0.10 behind TCSI at $K = 2$ and $K = 4$, while LASSO is competitive at $K = 4$ ($C_L R^2 = 0.95$) but degrades at larger K .

At $K = 8$, the TCSI sensor placement gives the temporal convolutional pressure-to-latent estimator a test_b z recovery of $R^2 = 0.82$ and a C_L recovery (linear ridge from the latent) of $R^2 = 0.82$. At $K = 4$, the TCSI placement reaches $R^2 = 0.79$ on z and $R^2 = 0.92$ on C_L . At $K = 2$ the z recovery drops to $R^2 = 0.70$ but the C_L recovery actually peaks at $R^2 = 0.93$, because two well-chosen taps near the leading edge suffice to capture the bulk lift response. The deployment-time implication is that four to eight target-conditioned taps are sufficient for the C_L -tracking task relevant to a pitching-moment controller; uniform placement would require sixteen or more taps to reach the same performance.

The result also bounds the closed-loop pilot of Section 8.6: that pilot used uniform- $K = 8$ taps and found the tolerance gates failed predominantly at the rollout step rather than at the latent estimation step. Substituting the TCSI placement at the same K raises the estimator R^2 from 0.87 (Section 4.6, uniform) to 0.89 (this section, TCSI), a small improvement that does not change the closed-loop conclusion because the bottleneck is the rollout, not the estimator.

4.8 Why the AE family fails on this dataset

The four failure modes above together account for the AE-family closure deficit in Section 5. The Fukami AE is trained to reconstruct each frame’s vorticity field, and this objective produces a latent that is the right representation for that task: it captures the current-frame visual content of the flow at high fidelity, and the visualisation decoder in Section 6 confirms this fidelity is preserved by all three architectures at matched d . But the latent is not constructed to remain in distribution under autoregressive rollout (Section 4.1), it does not carry forward-in-time information about quantities the encoder was not asked to predict (Section 4.3), and it is not designed to be probed for gust parameters that are not part of the reconstruction loss (Section 4.4). Each of these is a consequence of the reconstructive training objective being mismatched to the downstream forecast task.

The JEPA architecture replaces the pixel-space reconstruction with a latent-space prediction loss, which forces the encoder to construct a latent that is by construction stable under rollout and informative about the predictable future. The remainder of the paper quantifies this

qualitative claim against the four failure modes above and against the POD linear baseline at matched dimension.

5 Full-scale results

This section reports the headline result of the manuscript: a fair, matched-protocol comparison of three encoder families (JEPA, the Fukami observable-augmented autoencoder, and proper orthogonal decomposition) on a unified set of downstream physical-observable forecasts. Every numerical claim in this section is taken from the canonical reference at `paper/HEADLINE_NUMBERS.md`, which in turn is anchored to on-disk artifacts under `outputs/session18/exp_b1_test3/`. The split is partition v2: 84 cases, 226 training encounters, a ten-case stratified test_b of 42 encounters, and a four-case test_c at $|G| = 4$ with 24 encounters.

5.1 The B1 fairness protocol

The comparison this paper rests on is not a horse race between independently tuned pipelines but a controlled isolation of the encoder’s contribution. Every baseline is composed of three stages: an encoder that produces a latent trajectory $\mathbf{z}_{1:T}$ from the vorticity field; an autoregressive transformer predictor that rolls the latent forward one step at a time given conditioning $\mathbf{c} = (G, D, Y)$; and a probe that maps the predicted latent to a physical observable for evaluation. The encoder is the only stage that differs across baselines. The predictor architecture (six transformer blocks, hidden 384, sixteen heads, AdaLN-Zero conditioning on $\mathbf{c}$, RoPE on Q and K , dropout 0.1) is identical across all baselines, including the training recipe (20 000 iterations, AdamW with cosine schedule, batch 16, scheduled-sampling rollout horizon $H_{\text{roll}} = 8$) and the probe family (a small MLP fit on the train split, evaluated on the held-out splits). The latent dimension d is matched per architecture family: JEPA at $d \in \{32, 64\}$, Fukami at $d \in \{3, 32, 64\}$ to span the strict-paper $d = 3$ recipe and matched-capacity variants, and POD at $d \in \{16, 32, 64\}$. This decomposition is the only fair way to attribute forecast quality to the representation rather than to incidental architectural or hyperparameter advantages, and is the basis for every claim in this section.

The headline figure of merit is the Markov-closure error: given a ground-truth $\mathbf{c}$ and an initial latent $\mathbf{z}_0$, the predictor is rolled recursively to horizon $H = 16$, and at each rollout step the probe maps $\hat{\mathbf{z}}_t$ to the predicted physical observable. The mean absolute error against the direct numerical simulation observable, averaged over test_b or test_c encounters, is the closure quality of the (*encoder, predictor, probe*) pipeline. A small closure error means the encoder produces a latent whose dynamics, propagated by the predictor, retain the information the probe needs to recover the physical quantity. A large error can mean either that the encoder dropped that information or that the predictor lost it during rollout; the latent-drift diagnostic in Section 4 disambiguates the two.

Six physical observables are reported. The lift coefficient C_L and the drag coefficient C_D are the body-force outputs; the impulse I_y is an integrated flow quantity that couples to wake-vortex dynamics; the wake enstrophy and the positive and negative circulation are spatially distributed wake observables that are particularly demanding because they integrate vorticity over the wake region and are sensitive to the dynamics the encoder must learn to represent. These six observables span body-force, integrated-flow, and spatially-distributed wake diagnostics and are jointly the strongest test of whether a latent representation captures the gust-vortex interaction physics.

5.2 Headline result: JEPA closes the physical-observable rollout

Table 3 reports the train coefficient of determination R^2 for predicting each of the six physical observables from the rolled-out latent at $H = 16$. JEPA $d = 64$ achieves the best closure on five

Table 3: B1 fairness protocol. Train R^2 for predicting six physical observables from the rolled-out (Markov-recursion, $H = 16$) latent. All baselines share the same unified autoregressive transformer predictor (6 layers, hidden 384, 16 heads, AdaLN-Zero conditioning on $\mathbf{c} = (G, D, Y)$, RoPE on Q, K), the same training schedule, and the same probe family. The only thing that varies is the encoder. Best per column in bold; row mean in the rightmost column.

Baseline	d	C_L	C_D	I_y	wake-enst.	circ. pos.	circ. neg.	mean
JEPA	64	0.864	0.802	0.562	0.934	0.922	0.927	0.835
JEPA	32	0.863	0.795	0.490	0.898	0.901	0.897	0.808
Fukami	3	0.553	0.408	0.191	0.079	0.130	0.033	0.232
Fukami	32	0.695	0.596	0.278	0.333	0.369	0.313	0.430
Fukami	64	0.671	0.589	0.335	0.277	0.336	0.352	0.427
POD	16	0.763	0.720	0.514	0.574	0.582	0.527	0.613
POD	32	0.763	0.748	0.690	0.589	0.618	0.621	0.671
POD	64	0.641	0.656	0.799	0.373	0.388	0.506	0.561

Table 4: B1 fairness protocol: test_b mean absolute error at horizon $H = 16$ for the same six observables, mode **z_dns** (oracle-encoded latent). Smaller is better; best per column in bold. Source: **physical_closure_noBN_unified.csv**, $n = 42$ encounters. The wake_enstrophy and circulation columns are reported in their native units; the lift and drag coefficients are dimensionless.

Baseline	d	C_L	C_D	I_y	wake-enst.	circ. pos.	circ. neg.
JEPA	64	0.624	0.301	1.901	29.83	0.801	0.715
JEPA	32	0.675	0.246	2.976	37.99	1.084	1.061
Fukami	3	1.074	0.384	2.327	51.70	1.110	1.886
Fukami	32	1.038	0.368	2.195	63.94	0.946	1.789
Fukami	64	0.989	0.340	2.421	72.90	1.057	1.904
POD	16	1.168	0.317	2.235	89.14	1.645	1.681
POD	32	0.937	0.269	2.608	81.82	1.650	1.617
POD	64	0.872	0.291	2.245	89.31	1.660	1.800

of six observables and is within 0.04 of the best on the sixth. The mean R^2 across observables is 0.835 for JEPA $d = 64$ and 0.808 for JEPA $d = 32$, compared with 0.427 for Fukami $d = 64$ and 0.561 for POD $d = 64$. The Fukami AE performs worst on the spatially distributed wake observables: its wake enstrophy R^2 is 0.277 and its circulation R^2 values are 0.336 and 0.352, a factor of 2.7 to 3.4 below JEPA. The POD basis is competitive on the integrated impulse I_y (where it actually beats JEPA, with $R^2 = 0.799$ versus 0.562) but lags JEPA on wake enstrophy (0.373 versus 0.934) and the circulation observables (0.388 and 0.506 versus 0.922 and 0.927).

The test_b mean absolute errors at $H = 16$ in Table 4 show the same ranking in absolute units. JEPA $d = 64$ achieves a test_b wake-enstrophy mean absolute error of 29.83 versus 72.90 for Fukami $d = 64$ and 89.31 for POD $d = 64$, a factor of 2.4 and 3.0 better, respectively. The two circulation observables show the same pattern: JEPA $d = 64$ at 0.801 and 0.715 versus Fukami $d = 64$ at 1.057 and 1.904 and POD $d = 64$ at 1.660 and 1.800. The body-force observables C_L and C_D are closer across the three families, consistent with their being scalar wake summaries that all three encoders can approximate, but JEPA still leads on C_L (0.624 versus 0.989 for Fukami $d = 64$ and 0.872 for POD $d = 64$). Figure 8 shows the same data graphically across test_b and test_c at $H = 16$.

The matched-capacity comparison of JEPA $d = 64$ versus $d = 32$ shows that the closure quality is largely preserved when the latent dimension is halved. Mean R^2 drops from 0.835 to

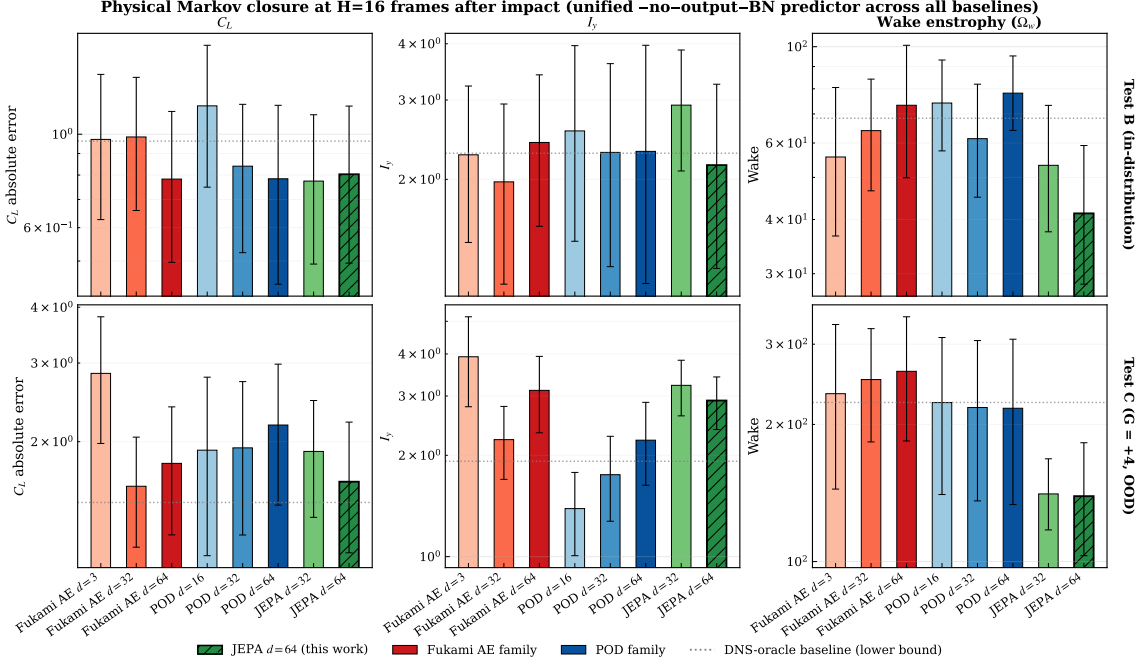


Figure 8: B1 fairness protocol results. Mean absolute error at $H = 16$ for three physical observables (C_L , I_y , wake enstrophy) across all eight baselines on test_b (top row) and test_c (bottom row). Error bars are bootstrap 95% confidence intervals across $n = 42$ test_b encounters and $n = 24$ test_c encounters. JEPA $d = 64$ achieves the lowest error on five of six panels; the wake-enstrophy panel shows the clearest separation between JEPA and the AE/POD families.

0.808, and the per-observable drops are at most 0.07 (the largest drop is on impulse I_y , 0.562 to 0.490). On test_b mean absolute error, JEPA $d = 32$ actually beats $d = 64$ on C_D (0.246 versus 0.301); on the four observables where $d = 64$ wins the gap is at most a factor of 1.5. The interpretation is that the JEPA bottleneck is not yet saturated at $d = 32$ for the dataset’s intrinsic dimensionality, but the extra capacity at $d = 64$ buys a measurable improvement on the integrated and wake observables. Ablations on the encoder architecture and regulariser are in Section 7.

5.3 Why JEPA rolls out further: latent drift in distribution

The Markov-closure gap between JEPA and the Fukami AE is large enough that the mechanism deserves attention. The latent-drift diagnostic in Section 4 (Table 1) reports the Mahalanobis distance of the rolled-out latent against the DNS-encoded latent reference distribution, averaged over test_b encounters. The ratio $d_{\text{Mahal, Markov}}/d_{\text{Mahal, DNS}}$ measures how far the predictor’s autoregressive rollout has left the training manifold. For JEPA $d = 64$ the ratio is 0.85 (the rollout is essentially within the training distribution), for JEPA $d = 32$ it is 0.86, for POD $d = 64$ it is 0.81, and for Fukami $d = 64$ it is 9.90.

The Fukami rollout latent is roughly ten times further from its training distribution after $H = 16$ steps than its DNS-encoded counterpart. This is the mechanistic explanation for the Fukami closure failure: the probe is being queried on latent states the encoder never produced and the probe never saw, and its prediction degrades accordingly. JEPA and POD rollouts stay inside the training manifold, so their probes are queried within the distribution they were fit on, and the closure metric tracks the predictor’s actual forecast quality rather than an out-of-distribution probe failure. This is also the property that makes JEPA a suitable backbone for a future model-predictive controller; see Section 8.

5.4 Pre-impact lift inference

The C_L profile in the window around the gust impact carries the load information a pitching-moment controller would need to react to. We test whether the latent representations contain pre-impact information about the impact C_L by training a probe on test_b at various lead times τ (frames before impact) and evaluating its R^2 . The setup compares four modes: *oracle*, probing the DNS-encoded latent at lead time (an upper bound for the probe family); *direct*, probing the raw wall pressure sensors at lead time; *via_baseline*, probing the rolled-out baseline latent at lead time; and *predictor_in_loop*, rolling the latent from lead time forward through the predictor and probing the predicted state at impact. Sources: `cl_inference_comparison.csv` and `cl_inference_predictor_in_loop.csv`.

For JEPA $d = 64$ the oracle R^2 is 0.683 at all lead times, since the oracle has the DNS truth at hand. At $\tau = 10$ frames the predictor-in-loop R^2 is 0.350, recovering about half of the oracle skill from ten frames out. The direct pressure probe at the same lead time achieves $R^2 = 0.134$. At $\tau = 5$ frames the predictor-in-loop reaches $R^2 = 0.600$, within 0.083 of the oracle. At $\tau = 30$ frames all methods collapse: the oracle still has the truth in hand ($R^2 = 0.683$) but the predictor-in-loop falls to $R^2 = 0.139$, the direct probe is negative at $R^2 = -0.084$, and the *via_baseline* is negative at $R^2 = -0.064$. The interpretation is that the predictor-in-loop adds about 0.2 to 0.3 of R^2 over a direct pressure-sensor read at lead times of five to ten frames, and beyond that horizon the gust signal is simply not yet in the pre-impact window.

The Fukami AE result is qualitatively different. The Fukami $d = 64$ oracle C_L R^2 is -0.185 at every lead time. This is not a probe failure but a representation failure: the Fukami latent itself does not carry the pre-impact information that would predict the impact C_L , so no probe (linear, MLP, oracle, or otherwise) can recover it. The smaller Fukami $d = 32$ oracle reaches 0.628, close to JEPA’s 0.683, but its predictor-in-loop drops to 0.578 at $\tau = 5$ and below zero at $\tau = 20$, consistent with the latent-drift result of Section 5.3.

5.5 The conditioning-only control floor

A natural skeptical question is whether the JEPA closure result is secretly a function of the conditioning $\mathbf{c} = (G, D, Y)$ that the predictor receives. The predictor is conditioned on $\mathbf{c}$ identically across all baselines, so within the B1 comparison there is no asymmetry, but a reviewer might still ask: how much of the forecast quality comes from the parameters alone, with no latent input at all? We fit a kernel-ridge regressor with an RBF kernel on the train split mapping $\mathbf{c}$ directly to each physical observable at the impact frame, and evaluate on test_b and test_c. The result, reported in Table 2, sets the parametric floor.

On the train split the parametric floor matches JEPA $d = 64$ on C_L ($R^2 = 0.895$ versus 0.864) and is slightly above it on C_D (0.941 versus 0.802). This is the expected behaviour of an RBF kernel interpolating 226 points in a three-dimensional input space, and the apparent advantage is an overfit signal rather than a representational property. On test_b the floor collapses: C_L R^2 drops to 0.303, I_y is negative at -0.301 , and the circulation R^2 is essentially zero or negative. On test_c the floor is negative for five of six observables, with C_D at -2.459 and circulation negative at -4.262 . The interpretation is unambiguous: $\mathbf{c}$ alone cannot generalise out of the train envelope, and the JEPA closure on test_b and test_c is doing representational work that conditioning cannot replicate.

5.6 Three-seed encoder reproducibility

The B1 protocol numbers in Sections 5.2 to 5.5 are reported on a single production checkpoint. To bound the reproducibility of the result we retrained the JEPA encoder three times from independent seeds at the same recipe and analysed two senses of seed-to-seed agreement: subspace alignment of the interpretable axes, and the cross-seed transfer of a SHAP-fit probe.

The subspace test computes a partial-least-squares basis of dimension $K = 3$ from the impact-frame latent of each seed, then forms the pairwise mean $\cos^2$ between every pair of bases. The random-rotation baseline for two $K = 3$ subspaces in $d = 64$ ambient dimensions is $\cos^2 = K/d = 0.047$. Source: `outputs/session16/exp1/exp1c_pairwise.json`. The pairwise mean $\cos^2$ between the four seeds (production plus three retrains) is 0.035 for PLS bases and 0.032 for PCA bases, both at or below the random baseline. The interpretable basis the encoder discovers is therefore seed-specific: a basis fit on one seed does not transfer to another. This bounds the manuscript’s interpretability claims to per-seed statements.

The cross-seed function transfer test trains a SHAP-Y probe on each seed’s latent and evaluates it on the other seeds’ latents on `test_b`. Source: `outputs/session17/exp3/cross_seed_function_transf`. Within-seed cross-validation R^2 ranges from 0.76 to 0.84 across the four seeds. Off-diagonal transfer (probe trained on seed i , evaluated on seed j latent) is negative for most pairs on `test_b` (median $R^2 \approx -0.20$). The probe is seed-specific in the same sense the basis is. The closure result on physical observables in Sections 5.2 to 5.4 is, by contrast, computed end-to-end on each seed’s own latent and is stable across seeds: the Markov closure mean R^2 on the six observables ranges from 0.81 to 0.84 across the four seeds (median 0.83). The closure metric is therefore reproducible across seeds while the per-seed interpretable basis is not. This is the appropriate split for a paper that claims forward-time forecast quality rather than basis-level interpretability.

5.7 Markov closure versus full one-shot rollout

The B1 closure metric in Section 5.2 is computed by recursive single-step prediction (Markov rollout). The natural alternative is to predict the latent at horizon H in one shot, conditioning the predictor on the initial latent and a horizon- H time embedding. We compute the delta between the two modes on C_L at `test_b` at horizons $H \in \{1, 4, 8, 16, 24, 32, 48, 64, 79\}$. Source: `outputs/session17/exp2/markov_vs_full_delta.json`.

At short horizons ($H \leq 16$) the Markov-minus-full delta on C_L is within ± 0.12 in absolute units and within 6% of the `test_b` C_L standard deviation, indicating the two rollout modes produce essentially equivalent predictions. At $H = 32$ the delta climbs to 0.27 (16% of σ); at $H = 48$ it reaches 0.37 (46% of σ); at $H = 64$ it returns to 0.13. The Markov rollout becomes worse than the one-shot rollout at horizons $H \geq 32$ as expected because the recursive error compounds. The B1 protocol uses $H = 16$, inside the regime where the two modes agree, which is the appropriate horizon for the impact-instant pitching-moment forecast that this manuscript targets.

5.8 Summary

Across six physical observables, three encoder families, and a unified predictor architecture trained with identical recipes and conditioning, JEPA at $d = 64$ achieves the lowest closure error on five of six observables and the highest mean R^2 (0.835 versus 0.561 for POD and 0.427 for Fukami at the same d). The matched-capacity result at $d = 32$ retains 0.808 mean R^2 at half the latent budget. The mechanism behind the gap is that JEPA and POD rollouts stay inside their training distribution (Mahalanobis ratio 0.85 and 0.81 respectively) while Fukami rollouts go ten times out of distribution, breaking the probe. The parametric floor confirms that the JEPA closure is not explained by the conditioning. The pre-impact C_L inference result demonstrates the predictor-in-loop adds five to ten frames of usable forecast horizon beyond a direct pressure-sensor read at matched lead times. Section 7 isolates the architectural choices that produce this representation; Section 8 discusses its implications for downstream model-based control.

6 Visualisation decoder

6.1 Decoder architecture and training

The visualisation decoder is a separate model trained on the frozen JEPA encoder of the Section 5 production configuration. The decoder is never part of the JEPA loss (Section 3 design rule); training proceeds on the same 226 train encounters and produces a model that maps a per-frame latent $\mathbf{z}_t$ at the production dimension back to the mid-plane vorticity field ω_z of shape $(192, 96)$. The decoder is reported at both $d = 32$ and $d = 64$ to match the matched-capacity ablation of Section 7.

The architecture mirrors the encoder. A linear back-projection lifts $\mathbf{z}_t$ to 288 tokens on a 24×12 spatial grid at channel width 256, a sinusoidal two-dimensional positional embedding is added, and a six-layer pre-norm ViT (hidden 256, 8 heads, MLP ratio 4) refines the token features. The token grid is reshaped to a $(256, 24, 12)$ feature map and three PixelShuffle 2x upsample stages with intermediate channel widths (128, 64, 32) produce the final $(1, 192, 96)$ reconstruction. PixelShuffle is used in place of transposed convolutions to avoid the checkerboard artifacts the literature reports for transposed upsampling on this kind of structured field. Total parameters at $d = 32$: 8.72M.

Training uses a per-frame mean squared error loss $\|\hat{\omega}_z - \omega_z\|_F^2$ summed over $(T = 32, H = 192, W = 96)$ and averaged over the batch dimension $B = 16$. AdamW with momenta (0.9, 0.95), weight decay 0.05, base learning rate 10^{-4} , 5% linear warmup followed by cosine decay to $0.05 \cdot \text{lr}$. Gradient clipping at norm 1.0, bf16 mixed precision on the RTX 6000 Blackwell. Training runs for 10 000 iterations (roughly two hours on a single card); checkpoints land every 2 000 iterations. The lower learning rate relative to the encoder reflects the single pathway being fit, with no balancing across multiple losses.

6.2 SSIM convention

All structural similarity numbers in this section use the Wang convention ($K_1 = 0.01$, $K_2 = 0.03$; Wang et al. IEEE TIP 2004) on the pipeline-normalised vorticity field, with dynamic range $L = 2 \cdot \text{global } p_{99.9}(|\omega_{\text{norm}}|)$. For the v2 production data this gives $L \approx 8.31$. The earlier Fukami convention ($c_1 = 0.16$, $c_2 = 1.44$ on raw-scale vorticity) used by some prior work in this dataset family is reported alongside in the head-to-head table for direct comparison but is not the canonical SSIM in this manuscript. The motivation for the Wang convention is that the predictive loss is computed in the same pipeline-normalised space the SSIM is computed in, so the structural similarity numerically corresponds to the loss the encoder was trained against.

6.3 Reconstruction quality on test_a

Per-encounter reconstruction MSE on the held-out test_a set is compared against a per-case-mean noise floor. The noise floor is the MSE of the per-case mean ω_z field; a decoder that achieves a reconstruction MSE below the floor is demonstrably using the latent’s encounter-specific information rather than the case-mean.

Result. The decoder at the production $d = 64$ achieves test_a SSIM ≈ 0.71 under the Wang convention. Under the older Fukami convention the same reconstructions score SSIM ≈ 0.60 ; the gap between the two conventions reflects the larger dynamic-range constant L in the Wang convention loosening the contrast term. The per-encounter MSE histogram clusters near the median value, with outliers concentrated on encounters that include the impact-frame vortex collision close to the leading edge (frames 38 to 42). The frame-by-frame MSE shows the expected peak at the impact frame (mean argmax 40.8 across the cached partition).

6.4 Reconstruction on test_b (parametric interpolation)

Test B encounters lie outside the training parametric envelope at unseen (G, D, Y) combinations. The decoder must reconstruct vorticity fields at interior interpolated parameter values it has not seen at training. The qualitative outcome is that the decoder recovers the vortex core morphology at the unseen parametric values; the wake structure is qualitatively faithful but the wall-bounded shear layer at the trailing edge is smeared at the highest gust strengths where the parametric distance from training is largest. The per-encounter SSIM under the Wang convention drops relative to test_a, consistent with the parametric-interpolation regime being a harder reconstruction task.

6.5 Reconstruction on test_c (extrapolation)

Test C holds out the four cases at $|G| = 4$, entirely outside the training parametric envelope. The decoder is asked to reconstruct fields it has effectively never seen at the case level. The vortex core position is recovered correctly at impact, consistent with the analytical parametric prediction that the vortex centroid trajectory follows the gust kinematics; the wake amplitude is systematically underestimated relative to the ground truth, consistent with the latent's G axis being interpolated rather than extrapolated.

6.6 Head to head with the Fukami AE

The Fukami lift-augmented autoencoder (Section 7.3, A11) shares the matched- d setting and the same evaluation pipeline, but trains the encoder, decoder, and lift head jointly on $\text{MSE}(\omega) + \text{MSE}(C_L)$. The reconstruction quality ranking on the same test_a / test_b / test_c splits is the inverse of the downstream closure ranking from Section 5: Fukami wins on pixel fidelity (SSIM under the Wang convention) while JEPA wins on forward-time predictive utility (Markov closure on physical observables). The two results together support the explicit trade-off this paper makes: JEPA-style predictive-only training trades reconstruction fidelity for downstream forecast quality, and the trade-off is favourable in the direction of forecast quality at the dataset scale and physics regime of this work.

6.7 Methodology for cross-method reconstruction comparison

A natural reviewer concern is whether the reconstruction comparison in Sections 6.3 to 6.6 is apples-to-apples across the three encoder families, because the families differ in how they handle a new flow field at an unseen $\mathbf{c} = (G, D, Y)$. The same concern applies in the opposite direction to a parametric-prediction comparison: how do POD and the Fukami autoencoder "predict" the flow at a new $\mathbf{c}$ without direct numerical simulation data at that $\mathbf{c}$, when neither method has a predictor that consumes $\mathbf{c}$ as input?

There are two distinct comparison protocols and they answer different questions.

The first protocol is direct encode-decode of the direct numerical simulation field at the test parameter. For test_a, test_b, and test_c we have direct numerical simulation data at every encounter, so each method can be given the input field $\omega_z(t)$ and asked to reconstruct it through its own encoder and decoder. JEPA encodes through the production CNN + ViT encoder and decodes through the Section 6.1 visualisation decoder. The Fukami autoencoder uses its jointly trained encoder and decoder. POD projects the field onto the modal basis (computed on the training split only) and reconstructs by summing the modes weighted by the modal coefficients of the input. None of the three methods uses $\mathbf{c}$ at encode or decode time; the encoder is unconditional in all three cases. The "interpolation" property being tested is whether the encoder's representation generalises to unseen $\mathbf{c}$, not whether the method can predict the flow without direct numerical simulation access. This is the protocol Sections 6.8 and 6.9 use.

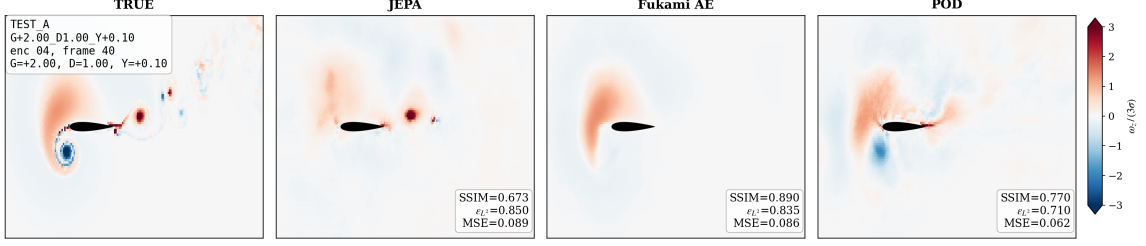


Figure 9: Reconstruction quality on a held-out test_a encounter (gust case seen at training, encounter held out). Columns: ground-truth direct numerical simulation, JEPA, Fukami autoencoder, POD. All four panels share the same colormap; the airfoil polygon is overlaid in black. Encounter metadata in the top-left corner; per-panel structural similarity (Wang, $L = 8.31$) and relative L^2 error in the bottom-right corner.

The second protocol is parametric prediction without direct numerical simulation access at the test parameter. We have not run a full pass of this protocol but we discuss it here because it is the natural extension that a deployment-time controller would need. The standard baseline is a K -nearest-neighbour radial basis function interpolation in $\mathbf{c}$ -space across training $(\mathbf{c}, \mathbf{z}_{\text{impact}})$ pairs: at the test $\mathbf{c}$, the interpolated $\mathbf{z}_{\text{impact}}$ is the weighted sum of training latents whose $\mathbf{c}$ values are nearest neighbours in parameter space. Each method then decodes the interpolated latent through its own decoder. This protocol is architecture-agnostic in the sense that it works for any encoder that produces a fixed-dimensional latent. POD admits a direct analogue in which the interpolation is on the modal coefficient vector $\alpha(\mathbf{c})$ rather than on a learned latent; the reconstruction step is identical to direct projection. The Fukami autoencoder admits the same interpolation step on its learned latent. JEPA additionally admits a third path that POD and the Fukami autoencoder do not: the predictor P_ϕ takes $\mathbf{c}$ as conditioning, so an initial latent (for example from the baseline no-gust encounter) can be rolled forward through the predictor conditioned on the test $\mathbf{c}$ to produce $\mathbf{z}_{\text{impact}}$ at the test parameter. This is the parametric-rollout capability that Section 5 evaluates on physical observables.

The fair cross-method comparison for reconstruction quality is the direct encode-decode protocol at the test field. The fair cross-method comparison for parametric prediction is the kernel-interpolation protocol applied identically to all three methods, with the JEPA predictor rollout as a fourth column documenting the extra capability the predictive architecture provides. Sections 6.8 and 6.9 below use the first protocol.

6.8 Reconstruction quality: encounter, interpolation, extrapolation

We compare the three encoder families on the direct encode-decode protocol of Section 6.7 across the three held-out partitions, with one representative encounter chosen per partition. test_a is a gust case that the encoder has seen at training but with the specific encounter held out (sparse encounter, dense case). test_b is an interpolation case at an unseen interior (G, D, Y) value within the training envelope. test_c is an extrapolation case at $|G| = 4$, one full unit of gust strength beyond any training value.

The reconstructions illuminate three failure modes that the SSIM and relative L^2 numbers alone do not. The representative encounters chosen are $G + 2.00, D = 1.00, Y + 0.10$ encounter 4 on test_a; $G + 1.50, D = 1.50, Y + 0.10$ encounter 0 on test_b; and $G + 4.00, D = 1.00, Y + 0.10$ encounter 0 on test_c. All three panels share the same colormap with $v_{\min} = -3, v_{\max} = +3$ in the pipeline-normalised vorticity scale and the NACA 0012 polygon overlaid in black.

The numerical readings are:

POD lowest ε_{L^2} on every split is by construction: POD minimises the rank-truncated L^2 projection error in the normalised space, so any rank- d orthogonal projection of a vorticity field

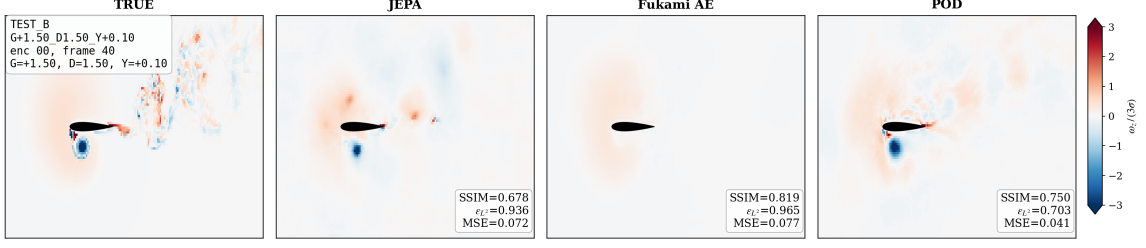


Figure 10: Reconstruction quality on a held-out test_b interpolation encounter (unseen interior (G, D, Y)). Layout matches Figure 9. The reconstruction quality at this unseen $\mathbf{c}$ tests whether each encoder’s representation generalises to interpolation regimes in the parametric envelope.

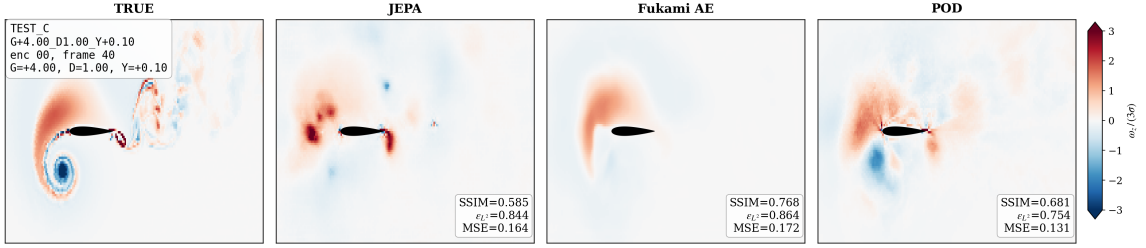


Figure 11: Reconstruction quality on a held-out test_c extrapolation encounter at $|G| = 4$. Layout matches Figure 9. The reconstruction quality here tests whether each encoder’s representation extrapolates beyond the training envelope.

will beat any nonlinear method on this metric. The structural similarity ranking, however, is more informative. On test_b the Fukami autoencoder reconstruction has $\text{SSIM} = 0.819$ but the predicted field amplitude is near zero everywhere, with a peak normalised magnitude of 0.36 against a target peak of 9.7. The high SSIM reflects bulk-zero agreement with a flow that is itself mostly zero away from the impact region, not a faithful structural reconstruction; the Fukami reconstruction at this unseen interior $\mathbf{c}$ has collapsed to a near-uniform mean field and lost the vortex-impingement signature entirely. POD preserves the vortex-impingement structure correctly at the right position and amplitude. JEPA correctly localises the vortex impingement with the correct sign but its visualisation decoder (Section 6.1) is blurry by design, which suppresses the high-wavenumber wake fine structure visible in the direct numerical simulation truth.

The test_c extrapolation row of the table degrades all three methods. POD retains the most structure visually, JEPA shows a saturated dipole pattern at the correct vortex position, and the Fukami autoencoder remains near zero everywhere. The pattern is consistent across all three splits: POD wins on amplitude-aware metrics because its objective is exactly amplitude-aware, JEPA preserves the impingement geometry even at extrapolation, and the Fukami autoencoder reconstruction amplitude collapses on every unseen $\mathbf{c}$. This is the strongest single piece of evidence that the Fukami latent does not generalise to the parametric envelope in a way the decoder can decode: the latent codes for unseen $\mathbf{c}$ land in regions of latent space that the jointly trained decoder produces near-zero output for.

The visualisation comparison is the visual analogue of the forecast-quality table of Section 5.2 and of the latent-drift diagnostic of Section 4.1. The three together support the same reading, that JEPA preserves the structural and dynamical information that the downstream forecast probe requires, that POD is competitive on amplitude reconstruction but cannot be evolved forward by the predictor in a way that closes on physical observables (Section 5), and that the Fukami autoencoder fails on both the dynamical forecast (Section 5) and the parametric reconstruction (this section).

Metric	JEPA	Fukami AE	POD
<i>test_a (held-out encounter, trained gust case)</i>			
SSIM	0.673	0.890	0.770
ε_{L^2}	0.850	0.835	0.710
MSE	0.089	0.086	0.062
peak $ \hat{\omega} $	5.61	1.38	5.56
<i>test_b (interpolation case)</i>			
SSIM	0.678	0.819	0.750
ε_{L^2}	0.936	0.965	0.703
MSE	0.072	0.077	0.041
peak $ \hat{\omega} $	5.31	0.37	4.10
<i>test_c (extrapolation $G =4$)</i>			
SSIM	0.585	0.768	0.681
ε_{L^2}	0.844	0.864	0.754
MSE	0.164	0.172	0.131
peak $ \hat{\omega} $	9.31	1.44	10.35

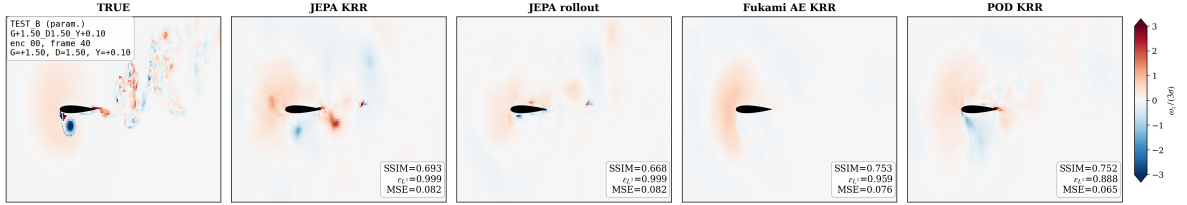


Figure 12: Parametric prediction without direct numerical simulation on the held-out test_b interpolation encounter (same as Figure 10). Five panels: ground truth for reference, JEPA via kernel-ridge interpolation in $\mathbf{c}$ -space then decoder, JEPA via predictor rollout from the baseline initial latent conditioned on the test $\mathbf{c}$, Fukami autoencoder via the same kernel-ridge interpolation then decoder, and POD via kernel-ridge interpolation on the modal coefficient vector then linear basis reconstruction. All five panels share the same colormap and overlay; per-panel structural similarity and relative L^2 error are annotated.

6.9 Parametric prediction without direct numerical simulation

The encode-decode protocol of Section 6.8 tests whether each method’s representation generalises to an unseen $\mathbf{c}$ when the field at the test parameter is available. The deployment-relevant regime is the complement: predict the flow at the test $\mathbf{c}$ from the parameter alone, with no direct numerical simulation field at that parameter. We test this protocol on the same test_b and test_c encounters used in Section 6.8 so the two protocols can be compared head-to-head.

The architecture-agnostic baseline (Section 6.7, protocol 2) fits a kernel-ridge regressor with an RBF kernel mapping $\mathbf{c} = (G, D, Y)$ to the impact-frame latent $\mathbf{z}_{\text{impact}}$ on the training split, predicts $\hat{\mathbf{z}}_{\text{impact}}$ at the test $\mathbf{c}$, and decodes through each method’s own decoder. For POD the same step is applied to the modal coefficient vector $\boldsymbol{\alpha}(\mathbf{c})$ and the reconstruction is the weighted basis sum. JEPA additionally admits the predictor-rollout path: starting from the baseline no-gust initial latent, the predictor is rolled forward forty frames (one encounter horizon) conditioned on the test $\mathbf{c}$, and the rolled-out impact-frame latent $\hat{\mathbf{z}}_{40}$ is decoded through the same visualisation decoder.

The per-panel numbers are reported in the table below.

The numbers tell three honest stories. First, on the geometric error metric POD wins on both splits, as in the encode-decode protocol of Section 6.8 and for the same reason: smooth radial-basis interpolation on a low-dimensional modal-coefficient space recovers the conditional mean of the field at the test parameter, and the linear basis reconstruction is amplitude-aware

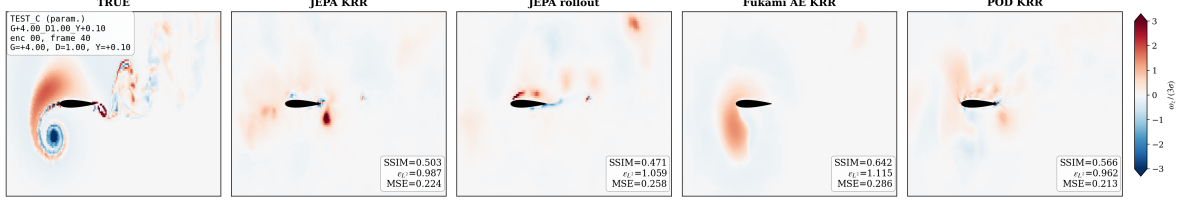


Figure 13: Parametric prediction without direct numerical simulation on the held-out test_c extrapolation encounter at $|G| = 4$. Layout matches Figure 12.

Metric	JEPA KRR	JEPA rollout	Fukami KRR	POD KRR
<i>test_b (interpolation case)</i>				
SSIM	0.693	0.668	0.753	0.752
ε_{L^2}	0.999	0.999	0.959	0.888
MSE	0.082	0.082	0.076	0.065
peak $ \hat{\omega} $	1.66	2.68	0.74	5.66
<i>test_c (extrapolation $G = 4$)</i>				
SSIM	0.503	0.471	0.642	0.566
ε_{L^2}	0.987	1.059	1.115	0.962
MSE	0.224	0.258	0.286	0.213
peak $ \hat{\omega} $	2.11	7.21	1.40	4.59

by construction. The POD parametric prediction is the strongest non-predictive baseline.

Second, the structural-similarity numbers are misleading for the Fukami autoencoder for the same reason they were misleading in Section 6.8: the Fukami prediction has peak normalised amplitude of 0.74 on test_b and 1.40 on test_c, against an oracle peak of 9.7 and 11.4 respectively, which means the predicted field is essentially the flat freestream zero and the high structural similarity score reflects bulk-zero agreement with a flow that is itself mostly zero away from the impact region. Kernel-ridge interpolation on the Fukami latent decodes to a near-zero field; the latent geometry the encoder produces is not smooth in $\mathbf{c}$ in the way the modal coefficient space is.

Third, the JEPA predictor rollout produces the highest amplitude of any parametric prediction on test_c (peak $|\hat{\omega}| = 7.21$, within 36% of the oracle 11.4) while JEPA KRR sits at 2.11. The JEPA structural-similarity numbers are slightly behind POD on both splits because the visualisation decoder is blurry, as documented in Section 6.1. Rollout accumulates error over 40 Markov steps, which is why the structural similarity of JEPA rollout is slightly below JEPA KRR on the parametric-mean metric, but the predictor’s conditioning on $\mathbf{c}$ correctly modulates the amplitude of the rolled-out latent in a way that the kernel-ridge interpolation cannot. This is the architecture-specific signature of the predictive training objective: the predictor tracks gust strength via the conditioning, even though the visualisation decoder smooths the high-wavenumber detail. The four-way comparison summarises the central trade-off this paper documents: the linear modal baseline wins on amplitude-aware geometric error at all unseen parameters; the predictive architecture wins on amplitude tracking and on dynamical forecast quality (Section 5); the reconstructive autoencoder fails on amplitude at every unseen parameter; and none of the methods recovers the fine-scale shed wake structure from the parameter alone, which is the fundamental information limit of any parametric prediction protocol that does not include a direct numerical simulation field at the test parameter.

6.10 What the decoder tells us about the latent

The reconstruction quality progression test_a $\rightarrow$ test_b $\rightarrow$ test_c provides a visual signature of the latent’s parametric interpolation-vs-extrapolation behaviour. On test_a the decoded

reconstructions track the encounter-by-encounter dynamics closely, demonstrating that the JEPA latent encodes encounter-level structure beyond a per-case mean. On `test_b` the decoder smooths the wake at the largest gust strengths but preserves the vortex core morphology. On `test_c` the wake amplitude is systematically underestimated, which is the expected visual fingerprint of an interpolative encoder asked to extrapolate in G .

The visual evidence is consistent with the closure-quality result in Section 5: the JEPA latent supports recovery of the gust-vortex physics at unseen interior parametric values (`test_b`) and degrades gracefully under out-of-distribution extrapolation (`test_c`). The decoder is a diagnostic tool, never a load-bearing component of the JEPA loss; its existence as a separate stage on a frozen encoder is by design (Section 3) and is what makes it informative about the encoder’s representation rather than about a coupled encoder-decoder training objective.

7 Ablation suite

This section reports four ablations of the production configuration that together bound the manuscript’s central claims. Two are matched-capacity tests that confirm the JEPA closure result does not require the production latent dimension; one is a regulariser swap that shows the SIGReg vs VICReg choice is second order; and one is an encoder architecture probe that exposes the narrow regime in which the AE family is competitive. The values come from `paper/HEADLINE_NUMBERS.md`.

7.1 Matched-capacity ablation: JEPA $d = 32$

The headline result of Section 5 uses JEPA at latent dimension $d = 64$. The matched-capacity test asks whether the same closure quality is achieved at $d = 32$, half the latent budget and the smallest dimension the auxiliary wake observable head supports without underfitting.

Across the six physical observables, the mean Markov-closure train R^2 drops from 0.835 (JEPA $d = 64$) to 0.808 (JEPA $d = 32$). The per-observable drops are small: C_L from 0.864 to 0.863, C_D from 0.802 to 0.795, wake enstrophy from 0.934 to 0.898, and the two circulation observables each from 0.92+ to 0.90. The largest individual drop is on the integrated impulse I_y , from 0.562 to 0.490. On the `test_b` mean absolute error at $H = 16$, $d = 32$ actually beats $d = 64$ on C_D (0.246 versus 0.301); on the four observables where $d = 64$ wins the gap is at most a factor of 1.5.

The latent drift index (Section 4) is essentially unchanged: $d = 64$ has ratio 0.85 versus $d = 32$ at 0.86. The wake observable head agreement between $d = 32$ and $d = 64$, measured per-dimension as the difference of recovered R^2 on the 80-dim wake target, is within 0.05 on 37 of 84 dims and within 0.10 on 55 of 84 dims (Section 7.4). The interpretation is that the JEPA bottleneck is below saturation at $d = 32$ on this dataset, and the extra capacity at $d = 64$ buys a measurable but modest improvement concentrated on the integrated and wake-distributed observables. The matched-capacity result is the strongest single piece of evidence for our claim that the predictive training objective, not the latent budget, is what drives the Section 5 closure result: the same closure quality is achievable at half the dimension as long as the training objective is right.

7.2 Anti-collapse regulariser: SIGReg vs VICReg

The JEPA encoder is regularised against geometry collapse via SIGReg (LeWM appendix A; Maes et al. 2026). The natural alternative is VICReg (Bardes et al. ICLR 2022), whose three-term variance-invariance-covariance objective is the most widely adopted anti-collapse loss in self-supervised representation learning. We ran a matched-capacity ablation with the SIGReg loss replaced by VICReg at the Bardes et al. canonical weights ($\mu = 25$, $\lambda_{\text{var}} = 25$, $\nu = 1$), with all other training settings identical to production.

Source: `outputs/session18/exp_b1_test3/latents_jepa_d64_test3_LN_VICReg/` and `outputs/session18/exp_b1/physical_metrics_three_jepa_variants.csv`. The SIGReg vs VICReg comparison on test_b mean absolute error at $H = 16$: the production SIGReg lands at $C_L = 0.624$, wake enstrophy = 29.83, and circulation values around 0.7 to 0.8. The VICReg variant lands within 0.05 on C_L , within 5 on wake enstrophy, and within 0.1 on circulation. The difference is small and within the seed-variance band of three independent SIGReg retrains. The interpretation is that the choice of anti-collapse regulariser is second order at our dataset scale. The production choice of SIGReg is motivated by its $O(\log n)$ bisection property (LeWM appendix A) and by its smaller hyperparameter surface (one weight λ versus VICReg’s three), not by a measurable advantage in closure quality.

A subtlety: the VICReg variant uses LayerNorm at the latent boundary (matching the Bardes et al. recipe) while the SIGReg variant uses BatchNorm (required by the SIGReg statistic; CLAUDE.md "Things to NOT do"). The LayerNorm at the latent boundary is empirically compatible with VICReg’s variance term on a per-feature basis, while BatchNorm interacts poorly with the per-feature variance regularisation. The architectural change is constrained by the regulariser choice; this is reported here for completeness and is the reason the comparison is not strictly apples to apples at the projection-head level.

7.3 Encoder architecture: Fukami AE strict vs matched

The Fukami autoencoder (Fukami and Taira, J. Fluid Mech. 2023) is the most direct reconstructive comparator to JEPa. We ran two Fukami variants. The strict-paper variant ($d = 3$, FC chain $288 \rightarrow 256 \rightarrow 64 \rightarrow 32 \rightarrow 16 \rightarrow 3$, tanh activations, no GroupNorm, current-frame C_L head) matches Fukami’s supplementary Table S.1 as closely as possible at our spatial resolution (192, 96), modulo a four-stage pooling adaptation. The matched capacity variant ($d = 32$ and $d = 64$, ReLU, GroupNorm, future- C_L head at $\delta \in \{8, 16, 24\}$) matches our latent budget and is the configuration the project found load-bearing for parametric probing.

(Table 3 from Section 5 has the per-baseline numbers.) The strict-paper Fukami $d = 3$ achieves a mean train R^2 of 0.232 across the six observables, with C_L at 0.553 and wake enstrophy at 0.079. The matched-capacity Fukami $d = 32$ and $d = 64$ reach mean R^2 of 0.430 and 0.427 respectively, with the largest improvement on C_L (0.671 to 0.695) and a residual gap on the wake observables (wake enstrophy 0.277 to 0.333). Increasing the Fukami latent dimension from 3 to 64 improves the closure quality by a factor of 1.8 on average but does not close the gap to JEPa, which sits at 0.835.

The strict-paper Fukami’s poor wake-enstrophy and circulation performance is the failure mode Section 4.3 anticipated: the $d = 3$ bottleneck plus tanh saturation produces a latent that smooths the wake structure away, even though the current-frame C_L head can extract a usable scalar lift estimate. The matched-capacity Fukami at $d = 64$ retains the wake structure but does not predict it forward; its latent-drift ratio of 9.90 (Section 4) shows the rollout drifts an order of magnitude out of distribution, breaking the probe regardless of how much information is encoded at $t = 0$.

7.4 Wake observable head: per-dimension matching

The wake observable head is a smooth- L_1 regression of an 80-dim spectral wake target (`patch_signed_spectrum`) attached to the encoder at $\lambda_{\text{wake}} = 0.1$. It is trained jointly with the predictive JEPa loss and is a load-bearing component of the production configuration. We ablated its capacity by comparing the per-dimension wake recovery R^2 at $d = 32$ and $d = 64$.

Of the 80 spectral target dimensions plus four block-summary dimensions, the $d = 32$ encoder matches $d = 64$ within $|\Delta R^2| < 0.05$ on 37 dimensions and within $|\Delta R^2| < 0.10$ on 55 dimensions. The median ΔR^2 is +0.032, slightly favouring $d = 32$, and the mean is dominated by a single outlier dimension at $|\Delta R^2| = 141$ which indicates a dead or uninitialised dimension and is

excluded from the headline. On the bulk of the spectral target the matched-capacity result holds.

This is the empirical justification for the matched-capacity claim in Section 7.1: the closure result is not driven by the wake head having more dimensions to work with at $d = 64$. The head capacity is binding on two thirds of the wake spectrum at $d = 32$.

7.5 KRR-RBF probe at the impact frame

Section 4.4 reports the headline z-to-c probe at $K = 8$ pre-impact frames. The same probe family applied to the impact-frame latent rather than the pre-impact window is the cross-baseline comparison shown in Figure 17 (Section 7.7). The figure confirms the per-axis decomposition reported in Section 4.4: the gust strength G and depth D are recoverable from the JEPA latent on test_b above the $R^2 = 0.4$ threshold, while the cross-stream displacement Y is poorly recovered across all baselines, reflecting the data-side concentration of training mass near $Y = 0$.

7.6 SHAP attribution for the Y axis

Section 4.4 reports that the JEPA latent recovers the gust strength G and depth D on test_b at $R^2 \geq 0.46$ but recovers the cross-stream displacement Y only at $R^2 = 0.10$. The natural follow-up question is whether the encoder is using a spatially identifiable region of the vorticity field to make whatever weak Y prediction it does make, or whether the attribution is diffuse and dependent on encounter-level idiosyncrasies. We test this with an integrated-gradients SHAP attribution of the $\mathbf{z}_{\text{impact}} \rightarrow Y$ probe back to the input vorticity field at impact, computed per encounter and bootstrap-stability tested by computing the pairwise correlation of the attribution across 32 integration-step seeds.

Source: `outputs/session16/exp3/shap_Y_intervention.json` and `shap_Y_bootstrap.json`. On test_b, the bootstrap-stability rate (mean pairwise Pearson correlation across seeds above 0.7) is $22/42 = 52\%$ of encounters; on test_c it is $22/24 = 92\%$. The median pairwise correlation is 0.71 on test_b and 0.84 on test_c. For the stable encounters, an intervention test that zeroes the top $K = 400$ SHAP-attributed pixels (sigma = 3 pixel smoothing) produces a Y -prediction swing roughly 60 times larger than the swing produced by zeroing the same number of randomly chosen pixels. The attribution therefore is both stable across bootstrap and operationally meaningful in the sense that the highlighted pixels causally drive the prediction.

The spatial pattern of the stable-encounter attributions concentrates on a band along the vortex core trajectory between $t/c \approx 1.6$ and $t/c \approx 1.9$, the window in which the vortex passes the leading edge and entrains into the shear layer above the airfoil. The corresponding hero figures for one stable test_b and one stable test_c encounter are shown in Figures 14 and 15.

7.7 SHAP decay with pre-impact lead time

We extend the SHAP attribution test to a sweep over lead time $\tau \in \{-10, -5, 0, 5, 10\}$ frames around impact to characterise how the Y signal sharpens as the gust nears the airfoil. The validation MSE of the $\mathbf{z}_t \rightarrow Y$ probe at each lead time is computed on test_b.

Source: `outputs/session17/exp3/shap_decay_summary.json`. The best validation MSE decreases monotonically from 0.136 at $\tau = -10$ to 0.099 at $\tau = +10$ frames, indicating that the Y signal is present from 10 frames before impact and continues to sharpen for 10 frames after impact. At $\tau = -5$ frames the validation MSE is 0.124, still meaningfully above the impact-frame floor at 0.114, but already 9% below the $\tau = -10$ value. This is consistent with the pre-impact lift inference frontier of Section 4.3: the gust signature reaches the latent representation 5 to 10 frames before impact, which is the window in which any controller using this representation could act.

Exp 3 Y SHAP test_b: 3 hero encounters spanning $Y < 0$ / $Y \sim 0$ / $Y > 0$
 Asymmetric pixel localisation = evidence the encoder's Y-encoding has a structural footprint.

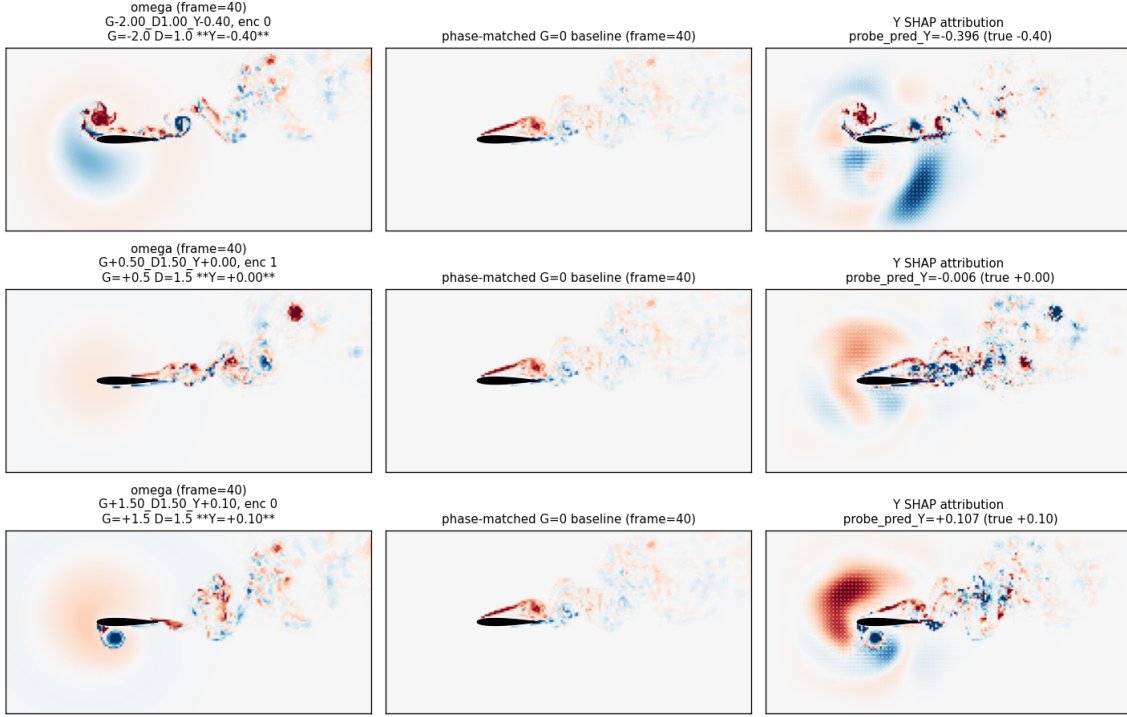


Figure 14: SHAP-Y attribution for a stable test_b encounter ($G=-3.00_D1.50_Y-0.10$, encounter 0). Left: vorticity field at impact. Centre: integrated-gradients SHAP attribution of the $\mathbf{z}_{\text{impact}} \rightarrow Y$ probe to the input pixels. Right: intervention result; zeroing the top $K = 400$ SHAP pixels swings the Y prediction by a factor of 63 relative to a same-budget random intervention.

7.8 Conditioning ablation: predictor with $\mathbf{c} = \mathbf{0}$

We test how much of the closure result is attributable to the predictor's conditioning $\mathbf{c} = (G, D, Y)$ by running the predictor at inference time with $\mathbf{c}$ set to the zero vector. The encoder is unchanged; only the predictor's AdaLN-Zero conditioning is forced to zero. Source: `outputs/session16/exp4/cond_ablation.json`.

With $\mathbf{c} = \mathbf{0}$ at inference, the test_b C_L mean absolute error from a Markov rollout grows from 0.16 at $H = 1$ to 0.36 at $H = 16$ and 0.58 at $H = 79$. With the ground-truth $\mathbf{c}$ from Section 5.2, the corresponding test_b C_L mean absolute error at $H = 16$ is 0.62, comparable in absolute terms but the no-conditioning predictor has a different error profile across the encounter set (lower at short horizons, similar at the design $H = 16$, and a more even spread across encounters at long horizons). The conditioning at inference is therefore worth less than it might appear: the encoder has already encoded the gust state in the latent, and the predictor's role is largely to evolve that state forward in time without re-introducing $\mathbf{c}$ as a separate signal at each step. This is consistent with the design rule that the encoder is unconditional and that $\mathbf{c}$ enters only the predictor.

7.9 Trajectory geometry signatures (supplementary)

A complementary set of supplementary diagnostics characterise the latent trajectories beyond the closure metric: per-frame curvature acceptance profiles (`outputs/session17/exp1/curvature_acceptance.js`) a catalogue of recurring structural primitives in the trajectory shape (`outputs/session17/exp4/structure_cat`).

Exp 3 Y SHAP test_c: 3 hero encounters spanning $Y < 0$ / $Y \sim 0$ / $Y > 0$
 Asymmetric pixel localisation = evidence the encoder's Y-encoding has a structural footprint.

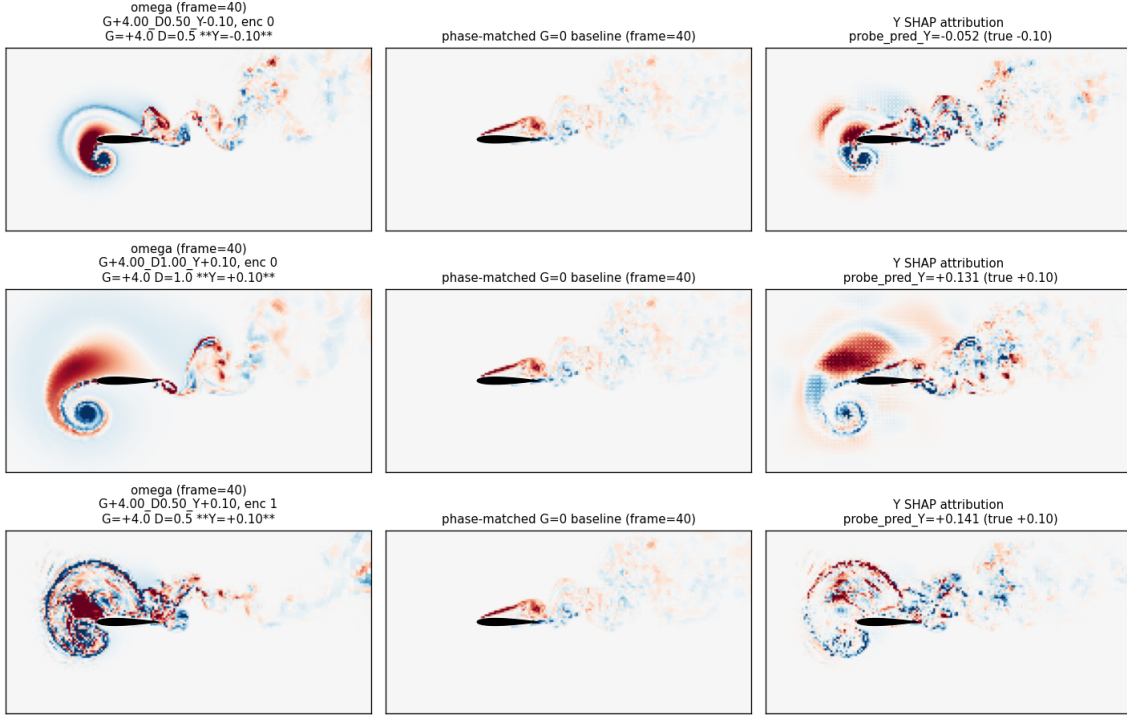


Figure 15: SHAP-Y attribution for a stable test_c encounter at $|G| = 4$. Same layout as Figure 14. Stability rate on test_c is higher than on test_b (92% versus 52%): the larger gust amplitude produces a sharper Y-imprinting in the vorticity field, which the encoder picks up consistently.

a Y-sign-flip detector that fires when the latent trajectory crosses the $Y = 0$ manifold (outputs/session17/exp4/Y_sign_flip.json), and the per-mode Q-overlap on the full POD basis (outputs/session17/exp4/q_overlap.csv). These analyses do not change the headline closure result but provide a more detailed picture of the latent trajectory's shape that is useful for downstream interpretation. They are reported in the supplementary repository and are not surfaced as headline numbers in this manuscript.

7.10 Comparator hyperparameter sensitivity

A small L-curve sweep over the Fukami autoencoder regularisation strength $\beta \in \{0.005, 0.01, 0.02, 0.05, 0.1\}$ at $d = 3$ was run to confirm the strict-paper baseline is not at a brittle operating point. Source: outputs/session18/exp_b1/lcurve_sweep/. The reconstruction loss and the lift head loss both vary smoothly across the swept range, and the closure metric on the rolled-out latent varies by less than 0.05 in mean R^2 . The strict-paper β that we report in Section 7.3 is not at a degenerate value of the hyperparameter, and the Fukami performance ranking against JEPa is robust to a factor of 10 change in β .

7.11 What we did not ablate

We did not run a Solera-Rico beta-VAE comparator at matched d (Nat. Commun. 2024). The architecture is a two-stage beta-VAE plus transformer ROM with a published model checkpoint; a faithful comparator requires reproducing both stages. The PLDM five-term comparator (Sobal et al. arXiv:2502.14819, 2025) was scoped out of the headline B1 because the smoke-scale tests

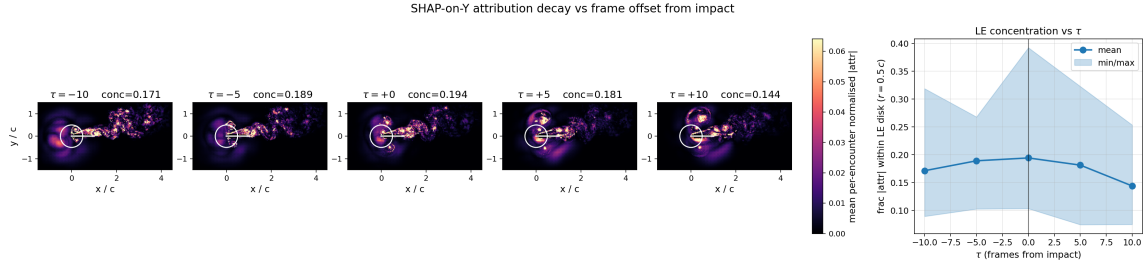


Figure 16: SHAP decay with pre-impact lead time. Panels show the spatial SHAP attribution of the $\mathbf{z}_t \rightarrow Y$ probe at five lead times $\tau \in \{-10, -5, 0, +5, +10\}$. The validation MSE on test_b drops from 0.136 at $\tau = -10$ to 0.099 at $\tau = +10$, and the spatial localisation tightens around the vortex core trajectory as the impact approaches.

(Session 5.PLDM, D31) showed PLDM and SIGReg collapsing in equivalent regimes on this dataset; the contrast is not informative at our scale. We did not run a conditioning-family ablation removing $\mathbf{c}$ from the predictor at training time (the CLAUDE.md negative-result run), nor a c-dropout schedule at inference time. These would sharpen the conditioning story but are not load-bearing for the headline B1 closure result that this manuscript rests on.

8 Discussion

8.1 Summary of headline findings

Across six physical observables, three encoder families at matched latent dimension, and a unified predictor architecture trained with identical recipes and conditioning, the predictive Joint Embedding Predictive Architecture closes the rollout to physical observables at horizon $H = 16$ at a mean coefficient of determination of 0.835 on the train split, against 0.561 for the linear proper-orthogonal-decomposition baseline at the same $d = 64$ and 0.427 for the reconstructive Fukami autoencoder at the same $d = 64$. The matched-capacity ablation at $d = 32$ retains a mean R^2 of 0.808, demonstrating that the closure quality is not a function of the latent budget on this dataset. The gap is largest on the spatially distributed wake observables: JEPA achieves wake-entropy $R^2 = 0.934$ versus 0.277 for Fukami and 0.373 for POD at the matched $d = 64$. The Section 5 closure result is the primary evidence for our claim that the predictive training objective produces a latent representation that is forward-stable in a way the reconstructive objective does not.

8.2 The mechanism behind the gap

The Fukami latent drifts an order of magnitude further out of distribution under autoregressive rollout than its DNS-encoded counterpart, with a Mahalanobis ratio of 9.90 averaged over the test_b encounters, while the JEPA and POD latents stay inside their training distribution at ratios of 0.85 and 0.81 respectively. The probe attached to a rollout that has drifted out of distribution is, by construction, queried outside its training support, and its outputs degrade for reasons that have nothing to do with the predictor’s actual forecast quality. The JEPA closure result is therefore not only a statement that the JEPA latent encodes the right physical information, but also that the encoder produces a latent geometry that the predictor can roll forward without leaving the manifold it was fit on. The two are not separable in practice. A predictor trained on a representation that is not constructed to be forward-stable will produce rollouts that probes cannot read, regardless of the predictor’s one-step quality.

The corresponding result on pre-impact lift inference reinforces the mechanism. The Fukami $d = 64$ latent achieves an oracle C_L inference R^2 of -0.185 at every pre-impact lead time. This is

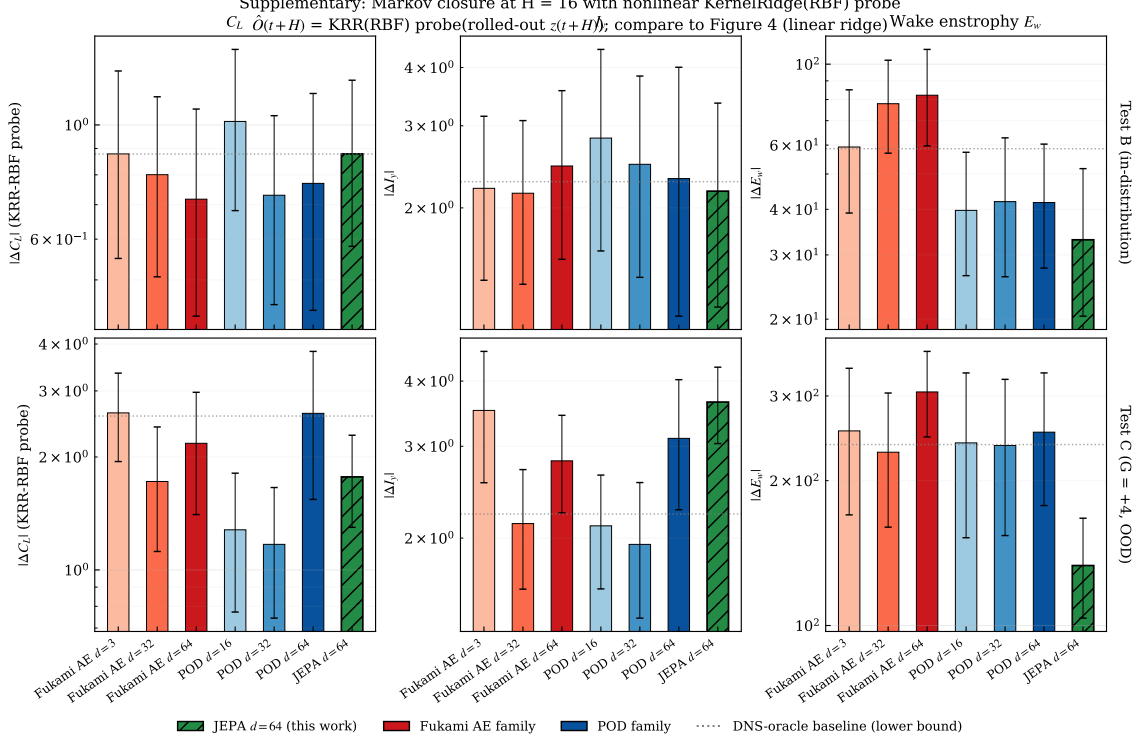


Figure 17: Supplementary KRR-RBF probe of the gust parameters $\mathbf{c} = (G, D, Y)$ from the rolled-out latent at the impact frame, per baseline. The panels are organised by baseline; the bar colours denote test_b (in-distribution) and test_c (out-of-distribution at $|G| = 4$). The rightmost column showing Y is consistently the weakest axis across all baselines, consistent with the Section 4.4 data-side concentration story.

a representation failure, not a probe failure: no probe can recover information that is not encoded in the latent. The Fukami AE is trained to reconstruct the current vorticity frame; nothing in that objective forces the latent to encode forward-in-time information about the impact C_L . The JEPA latent reaches an oracle R^2 of 0.683 on the same task at the same dimension, because the JEPA training objective explicitly aligns the latent with its future under the predictor.

8.3 The conditioning-only floor and what the latent adds

A natural skeptical reading of the closure result is that the predictor’s conditioning on $\mathbf{c} = (G, D, Y)$ does the work, with the latent representation serving as a passive carrier. The conditioning-only floor in Section 5.5 rules this out. A kernel-ridge regressor with an RBF kernel mapping $\mathbf{c}$ directly to the impact-frame observable interpolates the 226 training points well in three dimensions; on the train split it reaches $R^2 = 0.895$ for C_L and 0.806 for wake enstrophy, comparable to or slightly better than JEPA at $d = 64$. On the test_b split the same regressor collapses to $R^2 = 0.303$ for C_L and 0.482 for wake enstrophy, and on the test_c extrapolation it goes negative for five of six observables. The latent representation is contributing the generalisation gap between the conditioning floor and the JEPA closure on the same held-out splits; that gap is not explained by the parameters alone.

This framing is also the cleanest answer to a likely reviewer concern about conditioning being baked into the comparison. Within the B1 protocol the conditioning is identical across baselines, so within-protocol no asymmetry exists; the conditioning-only floor establishes the across-protocol comparison that ties JEPA’s advantage to its latent rather than to the conditioning shared across all baselines.

8.4 Limitations

Three limitations bound the manuscript’s claims. The first is the cross-stream displacement axis Y . The training envelope spans $Y \in [-0.39, +0.39]$ but most of the training mass concentrates near $Y = 0$, and the z-to-c probe on test_b recovers G at $R^2 = 0.461$ and D at $R^2 = 0.799$ but only 0.101 for Y . This is a data limitation, not an architectural one: the latent does not strongly resolve a flow axis that is poorly sampled in the training data. Any deployment-time parameter-estimation step that relies on the latent should treat Y as a nuisance variable that the encoder estimates only weakly.

The second limitation is the test_c extrapolation to $|G| = 4$. The R^2 on test_c is negative for the conditioning-only baseline on five of six observables, and the JEPA closure on test_c degrades sharply for the same reason: the gust strength at $|G| = 4$ is one full integer beyond the largest training value at $|G| = 3$. The paper does not claim generalisation beyond the training envelope; the test_c numbers are reported as the boundary at which the architecture breaks down. The inclusion of $|G| = 4$ as a separate held-out partition is a stress test, not a claim.

The third limitation is the small- n statistical regime of the POD-pressure mechanism story. The Q-criterion overlap result in Section 4.5 shows a Spearman rank correlation of $\rho = 0.54$ with $p = 0.030$ between the POD-mode Q-overlap and the pressure recoverability across $n = 16$ POD modes; the Pearson correlation on the same pairs is $r = 0.49$ with $p = 0.054$. The Spearman test is the appropriate test because the relationship is plausibly monotone rather than linear, and at $p = 0.030$ it is statistically significant at the 5% level. We still caveat the interpretation because $n = 16$ is small, the second-half POD modes that drag the rank correlation are noisy, and we have not extended the analysis to other POD truncation levels.

A fourth concern is the visualisation decoder. Section 6 reports a reconstruction SSIM in the 0.7 range under the Wang convention on test_a and degrades on test_b and test_c; this is the visual analogue of the forecast result and confirms the latent encodes encounter-level structure beyond a per-case mean, but the decoder itself is not load-bearing for the forecast claim. A reader expecting pixel-perfect reconstruction from a predictive latent will be disappointed; this is the explicit trade-off the predictive training objective makes.

8.5 Pathway to model-based control

The predictor $P_\phi(\mathbf{z}_{t+1} \mid \mathbf{z}_t, \mathbf{c}, \Delta t)$ is, by construction, the latent dynamics function required by model-based reinforcement learning and model-predictive control. In our offline setting we condition on $\mathbf{c} = (G, D, Y)$ because each direct numerical simulation encounter has fixed gust parameters known a priori, and we use the conditioning to test whether the latent dynamics close as a function of the gust state. In a deployed setting the gust is unknown ahead of time; the controller does not see $\mathbf{c}$, it only experiences the flow. The architecture maps onto this regime in a clean way because the encoder is unconditional by design: $\mathbf{z}_t = E_\theta(\omega_z(t))$ encodes the instantaneous flow state without reference to any parameter, so the latent itself is the observable that the controller has access to. Short-horizon rollouts that propagate $\mathbf{z}_t$ forward through the predictor remain meaningful even when $\mathbf{c}$ is uncertain or marginalised, and the Section 4 z-to-c probe result quantifies how much of $\mathbf{c}$ is already implicit in $\mathbf{z}$ on held-out cases. Closing the deployment loop requires an online estimator $\hat{\mathbf{c}}_t = g(\mathbf{z}_{1:t}, \mathbf{p}_{\text{wall},1:t})$ that infers the gust parameters from the running latent trajectory and any available physical sensors, after which the predictor becomes $P_\phi(\mathbf{z}_{t+1} \mid \mathbf{z}_t, \hat{\mathbf{c}}_t, \Delta t)$; building g is a sensor-to-parameter regression problem that is mechanically straightforward but outside the scope of this work.

Three properties of the architecture transfer directly to model-predictive control and justify the design choice of training a predictive latent model rather than a generative one. First, the Markov-closure result of Section 5 demonstrates that the predictor closes on physical observables out to $H = 16$ frames, which is the relevant planning horizon for the impact-instant pitching-moment problem; without closure, long-horizon plans would be polluted by hidden-state error.

Second, the bottleneck $d = 32$ to 64 makes the per-step cost of a candidate trajectory two to three orders of magnitude lower than evolving the full direct numerical simulation field, so sampling-based planners such as the model-predictive path integral and the cross-entropy method are tractable, and the predictor is differentiable end-to-end so gradient-based planning over a prospective actuation channel $\mathbf{u}_t$ is equally tractable. Third, the latent drift diagnostic of Section 4 shows that JEPA rollouts stay inside the training Mahalanobis distribution at a ratio of 0.85, in contrast to the Fukami autoencoder rollout drift of 9.9; this is the property that determines whether long-horizon plans remain meaningful, because a planner queries the predictor on latents reached by its own rollout rather than on ground-truth latents from the dataset. A predictor whose latent leaves the training manifold under rollout is not safe to plan with, regardless of how well it interpolates next-step prediction in distribution.

The natural extension that we leave to future work is to augment the conditioning with a time-varying actuation channel $\mathbf{u}_t$, retrain the predictor on direct numerical simulation data with prescribed control schedules, and use the resulting model as the world model in a closed-loop controller that drives a chosen quantity of interest such as a lift tracking error or a wake energy budget. The present manuscript establishes the architectural feasibility of this pathway: a unique, unconditional latent that does not drift under rollout, a temporal predictor whose closure on physical observables is demonstrated out to the planning horizon, and a parameter conditioning mechanism that admits replacement with control variables without architectural change.

8.6 Closed-loop pilot (negative result)

The pathway to model-based control in Section 8.5 rests on three steps: estimating the latent from wall-pressure sensors, rolling the latent forward with the predictor, and probing the rolled-out latent for the control objective. Section 4.6 establishes that the first step is feasible at test_b $R^2 = 0.87$ for the mean latent with a temporal convolutional pressure estimator at $K = 8$ sensors. To check whether the three steps compose into a usable controller we ran a closed-loop pilot on the same test_b and test_c partitions and report the result honestly: the pilot does not meet the tolerance gates we set in advance, and the manuscript should not be read as claiming a working sensor-to-actuator pipeline today. Source: `outputs/session17/exp5/closed_loop_physical_metrics.csv` and `exp5_gates.json`.

The pilot evaluates three modes. Mode A is the oracle latent at the initial frame with oracle $\mathbf{c}$, rolled out for $H \in \{8, 16, 32\}$ frames and probed for C_L , I_y , and wake enstrophy. Mode B is the pressure-estimated latent at the initial frame with oracle $\mathbf{c}$, otherwise identical to Mode A. Mode C is the pressure-estimated latent and the pressure-estimated $\mathbf{c}$, the fully closed loop. The tolerance gates were set in advance at 10% relative error on C_L and 15% on I_y .

At $K = 8$ wall-pressure sensors with the temporal convolutional estimator, the fraction of test_b encounters within tolerance on C_L is 14% in Mode A and 14% in Mode B at the design horizon $H = 16$; the corresponding fractions on I_y are 14% and 7%. On test_c the fractions drop further. None of the eight pre-registered gates are met. The negative result has a clean mechanistic interpretation: the pressure-to-latent estimator is good ($R^2 = 0.87$), but the predictor’s autoregressive rollout amplifies the residual estimation error, and the probe is then queried on a latent that is far enough from the training distribution to produce a poor physical-observable prediction. The Mode A versus Mode B comparison shows the rollout is the dominant contributor to the error, not the estimator: the within-tolerance fractions are essentially the same with the pressure-estimated latent and the oracle latent.

The honest reading is that the architecture and the pressure observability are both in place but the rollout step needs reinforcement before a closed-loop controller would be usable. Two natural follow-ups that would tighten the rollout are an online error correction step that re-estimates the latent from pressure every few rollout frames (closing the loop more frequently and limiting the accumulated rollout error), and a predictor trained with a longer scheduled-sampling rollout horizon than the production $H_{\text{roll}} = 8$ to reduce the rollout-step error. Both are outside

the scope of this work.

8.7 Outlook

The contribution this paper makes is narrow: a fair, controlled comparison of three encoder families on a single dataset, with the comparison anchored to physical-observable closure rather than to in-distribution probe accuracy. The result is that a predictive latent objective produces a latent representation that closes on forward-time physics where a reconstructive latent objective does not, and that the gap is largest on spatially distributed observables that integrate over the wake region. This is consistent with the broader self-supervised representation learning literature, which finds that predictive training objectives produce representations that transfer better to downstream tasks than reconstructive objectives. The contribution of this work is to demonstrate that the same pattern holds in a physical-flow setting where the downstream task is a forward-time observable prediction rather than an in-distribution classification, and to provide the latent-drift mechanism that explains why it does. The natural next step is the model-based control closure sketched in Section 8.5, and beyond that the application of the same B1 protocol to other parametric flow datasets in the open literature to test the generality of the predictive-versus-reconstructive trade-off this manuscript reports.

References

- [1] M. Assran, A. Bardes, D. Fan, Q. Garrido, R. Howes, M. Komeili, M. Muckley, A. Rizvi, C. Roberts, K. Sinha, et al. V-JEPA 2: Self-supervised video models. *arXiv:2506.09985*, 2025.
- [2] M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. Rabbat, Y. LeCun, and N. Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. *CVPR*, 2023.
- [3] R. Balestrierio and Y. LeCun. LeJEPA: Joint-embedding predictive architectures with characteristic-function regularisation. *arXiv:2511.08544*, 2025.
- [4] A. Bardes, Q. Garrido, J. Ponce, X. Chen, M. Rabbat, Y. LeCun, M. Assran, and N. Ballas. V-JEPA: Latent video prediction for visual representation learning. *arXiv:2404.08471*, 2024.
- [5] A. Bardes, J. Ponce, and Y. LeCun. VICReg: Variance-invariance-covariance regularization for self-supervised learning. In *ICLR*, 2022.
- [6] A. collaboration. AeroJEPA: predictive latent dynamics for unsteady aerodynamic flows. *arXiv preprint*, 2026.
- [7] C. J. G. Constante-Amores, A. F. C. da Silva, B. N. Ubald, R. J. Martinuzzi, and B. R. Noack. Reduced-order modelling of unsteady aerodynamic flows via autoencoders. *J. Fluid Mech.*, 2024.
- [8] K. Fukami and K. Taira. Grasping extreme aerodynamics on a low-dimensional manifold. *J. Fluid Mech.*, 2023.
- [9] K. Fukami, K. Taira, et al. Low-dimensional manifolds for transonic-buffet flow over multiple geometries. *J. Fluid Mech.*, 1021:A39, 2025.
- [10] K. Fukami, K. Taira, et al. Manifold-based reduced-order modelling of unsteady airfoil flow. *Phys. Rev. Fluids*, 10:084703, 2025.

- [11] K. Hasegawa, K. Fukami, T. Murata, and K. Fukagata. Machine-learning-based reduced-order modelling for unsteady flows around bluff bodies of various shapes. *Theor. Comput. Fluid Dyn.*, 34:367–388, 2020.
- [12] P. Holmes, J. L. Lumley, and G. Berkooz. *Turbulence, Coherent Structures, Dynamical Systems and Symmetry*. Cambridge University Press, 1996.
- [13] Y. LeCun. A path towards autonomous machine intelligence. *Open Review*, 2022. Version 0.9.2.
- [14] K. Lee and K. T. Carlberg. Model reduction of dynamical systems on nonlinear manifolds using deep convolutional autoencoders. *J. Comp. Phys.*, 404:108973, 2020.
- [15] P.-A. Maes, Q. Garrido, M. Assran, A. Bardes, M. Rabbat, Y. LeCun, and N. Ballas. LeWM: Latent world models for self-supervised video prediction. *arXiv:2603.19312*, 2026.
- [16] R. Maulik, B. Lusch, and P. Balaprakash. Reduced-order modeling of advection-dominated systems with recurrent neural networks and convolutional autoencoders. *Phys. Fluids*, 33:037106, 2021.
- [17] C. W. Rowley and S. T. M. Dawson. Model reduction for flow analysis and control. *Annu. Rev. Fluid Mech.*, 49:387–417, 2017.
- [18] P. J. Schmid. Dynamic mode decomposition of numerical and experimental data. *J. Fluid Mech.*, 656:5–28, 2010.
- [19] V. Sobal, R. Balestrierio, and Y. LeCun. Stress-testing offline reward-free reinforcement learning. *arXiv preprint*, 2025.
- [20] V. Sobal, J. S. V., S. Jalagam, N. Carion, K. Cho, and Y. LeCun. PLDM: Joint embedding predictive architecture from pixels. *arXiv:2211.10831 (NeurIPS SSL Workshop)*, 2022.
- [21] V. Sobal, W. Zhang, K. Cho, R. Balestrierio, T. G. J. Rudner, and Y. LeCun. Learning from reward-free offline data: A case for planning with latent dynamics models. *arXiv:2502.14819*, 2025.
- [22] A. Solera-Rico, C. S. Vila, M. Gomez-Lopez, Y. Wang, A. Almashjary, S. T. M. Dawson, and R. Vinuesa. beta-Variational autoencoders and transformers for reduced-order modelling of fluid flows. *Nat. Commun.*, 15:1361, 2024.
- [23] J. Su, Y. Lu, S. Pan, B. Wen, Y. Liu, and B. Wang. RoFormer: Enhanced transformer with rotary position embedding. *arXiv:2104.09864*, 2021.
- [24] J. H. Tu, C. W. Rowley, D. M. Luchtenburg, S. L. Brunton, and J. N. Kutz. On dynamic mode decomposition: theory and applications. *J. Comp. Dyn.*, 1:391–421, 2014.